

# Microeconomic Theory I

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Fall 2026 edition

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# Preface to the Fall 2026 Edition

These notes provide a rigorous first course in graduate microeconomic theory. The Fall 2026 edition preserves the substantive coverage of the Fall 2024 course while improving the exposition, notation, organization, and portability of the source files. The central discipline throughout is to distinguish primitives, behavioral assumptions, optimization problems, equilibrium conditions, and empirical implications.

The material is cumulative. Part I develops the mathematical language of binary relations and connects preferences to observable choice. Part II derives individual demand and extends the benchmark model to limited attention and uncertainty. Part III studies aggregation and stochastic choice. Part IV develops production, competitive markets, and general equilibrium. Part V treats equilibrium restrictions, trade under uncertainty, matching, and computable general equilibrium.

# Notation and Conventions

The following conventions apply unless a chapter explicitly states otherwise. They are collected here so that the same symbol does not silently change meaning across topics.

- There are  $L$  commodities, indexed by  $\ell = 1, \dots, L$ . Commodity bundles are column vectors  $x \in X \subseteq \mathbb{R}_+^L$ .
- Consumers are indexed by  $i = 1, \dots, I$ ; firms are indexed by  $j = 1, \dots, J$ . Consumption is denoted by  $x$ , production by  $y$ , and endowments by  $\omega$ .
- Prices are column vectors  $p \in \mathbb{R}_{++}^L$ , unless a result explicitly permits zero boundary prices. Set difference is denoted by  $A \setminus B$ . The operators  $\max$  and  $\min$  are typeset as mathematical operators. The scalar  $w > 0$  denotes wealth. The budget set is  $B(p, w) = \{x \in X : p \cdot x \leq w\}$ .
- Marshallian demand is  $x(p, w)$ , Hicksian demand is  $h(p, \bar{u})$ , indirect utility is  $v(p, w)$ , and the expenditure function is  $e(p, \bar{u})$ .
- The dot  $\cdot$  denotes the Euclidean inner product, and  $\top$  denotes transpose. A prime marks an alternative object, as in  $p'$  or  $x'$ ; it is not used for transpose.
- Weak preference, strict preference, and indifference are denoted by  $\succeq$ ,  $\succ$ , and  $\sim$ , respectively. Choice rules are denoted by  $c$  or  $C$  only when the surrounding section defines the domain and whether the rule is single- or set-valued.
- For a differentiable vector-valued function,  $D_p x$  is its Jacobian with respect to prices, while  $\nabla u$  is the column gradient of a scalar-valued function.

## Part I

# Foundations of Choice and Demand

# Chapter 1

## Basics of Consumer Theory

### 1.1 Preliminaries

Consumer theory begins with feasible choice and observable demand. This chapter introduces the commodity space, budget sets, demand, Walras' law, homogeneity of degree zero, the weak axiom of revealed preference, and the Slutsky matrix. The next chapter develops the abstract connection between choice and preference; the utility-maximization chapter then joins these two perspectives.

### 1.2 Environment: Commodities and Budget Set

First we describe formally the goods and services that a consumer may consume. We call the object of consumption **commodity**. We assume that the **commodity space** is  $\mathbb{R}^L$ , where  $L \geq 1$  is the number of commodities. We refer to any commodity by  $l \in \{1, \dots, L\}$  and to the quantity of the commodity by  $\bar{x}_l$ . The **consumption bundle or commodity bundle** is given by the vector  $\bar{x} \in \mathbb{R}^L, \bar{x} = (\bar{x}_1, \dots, \bar{x}_L)^\top$ .

Because the commodity space may be too generic for some modeling purposes (e.g., the situation at hand may rule out extremely large quantities of some commodities, or negative quantities are impossible) we use from now on the **consumption set**  $X \subseteq \mathbb{R}^L$ . Remember that the consumption set is an abstract concept that can be defined according to each modeling situation, but from now on we will let  $X = \mathbb{R}_+^L$ .

In real life, the consumer cannot consume anything he likes from the consumption set  $X$ . He is constrained either by his available wealth or by other reasons. Here we focus on cases

where the only constraint is wealth. Let  $w > 0$  be the consumer's wealth (broadly defined). Let  $p \in \mathbb{R}_{++}^L$  be the vector of commodity prices. The consumer is then limited to choose from his budget set. The budget set is a subset of the choice set  $B(p, w) \subseteq X$  and it is defined by  $B(p, w) = \{x \in X : p \cdot x \leq w\}$ .

### 1.3 Demand Functions

After specifying the consumer environment, we identify the **primitive** to be modeled: the demand function. The demand function is a mapping  $x : \mathbb{R}_{++}^L \times \mathbb{R}_{++} \rightarrow X$  ( $(p, w) \rightarrow \bar{x}$ ) that takes as inputs price-wealth pairs and produces a commodity bundle such that  $x(p, w) \in B(p, w)$  (i.e.,  $p \cdot x(p, w) \leq w$ ).

In fact  $x_l(p, w)$  denotes the quantity of commodity  $l$  consumed at prices and wealth  $(p, w)$ .

Observe that  $x$  is an empirical object, we have not explained yet how it came to be. Also note that we have made a very strong assumption when taking a function as our primitive. Observing a demand function would require observing the consumer's choice at every price-wealth pair, which is impossible in practice. We therefore use finite observations and testable restrictions throughout the book.

**Example 1.** Cobb-Douglas Demand Function.  $x(p, w) = \left(\frac{\alpha w}{p_1}, \frac{(1-\alpha)w}{p_2}\right)^\top$  for  $\alpha \in (0, 1)$ .

### 1.4 Consistency in Consumption

With the environment and demand primitive in place, we turn to consistency in consumption choices. This approach restricts observable behavior before imposing a specific preference representation.

We will impose a set of restrictions or axioms over the Marshallian demand function that we expect to hold in all situations irrespective of the details. Here we understand modeling consumption as restricting the set of allowable behaviors.

These axioms are minimal theoretical requirements, not empirical laws that every dataset must satisfy. Their value is that they make the set of admissible behavior explicit and provide a baseline against which richer models can be compared.

**Axiom 1.** (*Walras' law*) A demand function  $x$  is said to satisfy Walras' law if and only if  $p \cdot x(p, w) = w$  for all price-wealth pairs.

Walras' law imposes budget exhaustion; it rules out choices that leave unspent wealth when a locally nonsatiated consumer could purchase more.

**Axiom 2.** (*No money illusion/Homogeneity of degree zero (HD0)*) A demand function  $x$  is said to satisfy Homogeneity of Degree Zero if and only if for any  $\alpha > 0$   $x(\alpha p, \alpha w) = x(p, w)$ .

HD0 says that multiplying all prices and wealth by the same positive scalar does not change demand. It is a normalization of units, not a behavioral claim about intelligence.

**Axiom 3.** (*Weak Axiom of Revealed Preference -WARP*) A demand function  $x$  is said to satisfy the WARP if and only if for any two distinct price-wealth pairs  $(p, w)$  and  $(p^*, w^*)$ , and  $x(p, w) \neq x(p^*, w^*)$  it follows that if  $p \cdot x(p^*, w^*) \leq w$  then  $p^* \cdot x(p, w) > w^*$ .

The WARP is the most interesting of the restrictions presented until this point. It captures the idea that there is stability in the choices that consumers make. Notice the 'revealed preference' in the name of the WARP, it comes from the idea that by observing demand behavior underlying consumer preferences are revealed to the observer.

Let's define a revealed preference relation  $R^D$  that stands for "directly revealed preferred to", this is a binary relation on the consumption set  $X$  (i.e.,  $R^D \subseteq X \times X$ ) defined as follows:

Let  $(p^1, x^1)$  and  $(p^2, x^2)$  be observed consumption bundles with its corresponding price, then

$$x^1 R^D x^2 \iff (p^1)^\top x^1 \geq (p^1)^\top x^2.$$

Notice that the hypothesis in the WARP says  $p \cdot x(p^*, w^*) \leq w$ , let  $y = x(p^*, w^*)$  and recall that by Walras' law  $p \cdot x(p, w) = w$ , also let  $z = x(p, w)$ . This means that  $w = p \cdot z \geq p \cdot y$  which implies that  $z R^D y$ , or in words,  $z$  is directly revealed preferred to  $y$ . The WARP is then the requirement that if  $z R^D y$  then  $y$  cannot be strictly directly revealed preferred to  $z$ . This means that we do not allow  $p^* \cdot z \leq p^* \cdot y$  or equivalently it must be the case that  $p^* \cdot x(p, w) > w^*$ . The binary relation  $R^D$  is declared to satisfy the weak axiom. Recall that this relation is not necessarily representing actual tastes/preferences, we do not need them to do that. We are only establishing a condition that for many economists (back in the time of Samuelson) was the minimal requirement that all consumers fulfilled.

## 1.5 Comparative Statics and Slutsky Matrix

After establishing the axioms of behavior, the next step is to study their implications for the parameters of the model. The parameters of the model in our study of consumer behavior are

prices and wealth pairs.

- The price Jacobian is  $D_p x(p, w) \in \mathbb{R}^{L \times L}$ ; its  $(\ell, k)$  entry is  $\partial_{p_k} x_\ell(p, w)$ .
- We let  $D_w x(p, w) = (\partial_w x_\ell(p, w))_{\ell=1}^L$  be the vector of wealth effects.
- The Slutsky matrix of a demand function  $x$ , is  $S(p, w) = D_p x(p, w) + D_w x(p, w)x(p, w)^\top$  or the total change of demand with respect to a “Slutsky compensated variation of prices”.  
A Slutsky-compensated price variation leaves the previously chosen bundle affordable at the new prices.  $S_{\ell k} = \partial_{p_k} x_\ell(p, w) + \partial_w x_\ell(p, w)x_k$ .

Fix a reference bundle  $q$  and consider the compensated demand  $x(p, p \cdot q)$ . The chain rule gives  $D_p[x(p, p \cdot q)] = D_p x(p, p \cdot q) + D_w x(p, p \cdot q)q^\top$ , when  $q = x(p, w)$  and Walras’ law gives  $p \cdot q = w$ , this derivative is the Slutsky matrix:  $S(p, w) = D_p x(p, w) + D_w x(p, w)x(p, w)^\top$ .

### Implications of Walras’ Law for the Slutsky Matrix

**Fact 1.** Let  $x$  satisfy Walras’ law then: (i)  $p^\top D_w x(p, w) = 1$ , (ii)  $p^\top D_p x(p, w) = -x(p, w)^\top$ , (iii)  $p^\top S(p, w) = 0$ .

The previous fact part (i) implies that there exists at least one normal good, the reason is that there has to be at least one  $l$  such that  $\partial_w x_l(p, w) > 0$  for  $\sum_\ell p_\ell \partial_w x_\ell(p, w) = 1$ . In general, Walras’ law restricts the wealth effects, while part (iii) restricts consumption changes under Slutsky-compensated price changes.

### Implications of Homogeneity of Degree Zero.

*Claim 1.* Let  $x$  satisfy Walras’ law and homogeneity of degree zero then  $S(p, w)p = 0$ .

*Proof.* By HD0 we have that  $x(\lambda p, \lambda w) = x(p, w)$  for all  $\lambda > 0$ , we derive by  $\lambda$  and obtain.

$D_p x(\lambda p, \lambda w)p + D_w x(\lambda p, \lambda w)w = 0$ . We let  $\lambda = 1$ , and recall that  $p \cdot x(p, w) = w$  by Walras’ law then:

$$D_p x(p, w)p + D_w x(p, w)p \cdot x = S(p, w)p.$$

□

### Implications of WARP.

The main implication of the WARP while assuming Walras’ law is the Compensated Law of Demand (CLD).

**Definition 1.** (CLD) We say that  $x$  satisfies the Compensated Law of Demand if and only if  $(p^* - p) \cdot (x(p^*, w^*) - x(p, w)) < 0$  when  $x(p^*, w^*) \neq x(p, w)$  and  $w^* = p^* \cdot x(p, w)$ .

We next establish that WARP implies the compensated law of demand.

**Proposition 1.** *Let  $x$  satisfy Walras' law and WARP. Then  $x$  satisfies the CLD.*

*Proof.* Let  $z = x(p, w)$  and  $y = x(p^*, w^*)$ .

By the assumption that  $p^* x(p, w) = w^*$  and by Walras' law we have that  $p \cdot z = w$ , and  $p^* \cdot z = w^*$ . Also we have  $p^* \cdot y = w^*$ .

Thus  $p^* \cdot (y - z) = 0$ . By WARP since  $p^* \cdot z \leq w^*$  it must be the case that  $p \cdot y > w$ .

We have then  $p \cdot y - p \cdot z > 0$  or  $p \cdot (y - z) > 0$  subtract this from that  $p^* \cdot (y - z) = 0$  to obtain:

$$(p^* - p) \cdot (y - z) < 0,$$

$$y \neq z.$$

□

The differential version yields the following corollary.

**Corollary 1.** *Let  $x$  satisfies Walras' law and WARP. Then  $x$  is such that the symmetric part of the Slutsky matrix,  $S(p, w)$ , is negative semidefinite: for every  $a \in \mathbb{R}^L$  and every price-wealth pair,  $a^\top S(p, w) a \leq 0$ .*

*Proof.* If  $x$  satisfies Walras' law and WARP. Then,

$$(p^* - p) \cdot [x(p^*, p^* \cdot x(p, w)) - x(p, w)] \leq 0.$$

I consider a case such that  $p^*$ , such that  $dp = (p^* - p)$ .

The differential version of the CLD is:

$$dp \cdot dx \leq 0.$$

I compute  $dx = D_p x(p, w) dp + D_w x(p, w) dw$ , notice that by Slutsky compensation:

$$dw = x(p, w) \cdot dp = dp \cdot x(p, w).$$

$$dp \cdot dx = dp^\top [D_p x(p, w) + D_w x(p, w) x(p, w)^\top] dp,$$

$$dp \cdot dx = dp^\top S(p, w) dp \leq 0.$$

□

### Marshallian and compensated price effects

The Slutsky decomposition separates the total response of demand to a price change from the response at fixed purchasing power. For a differentiable demand function,

$$\partial_{p_k} x_\ell(p, w) = S_{\ell k}(p, w) - x_k(p, w) \partial_w x_\ell(p, w).$$

Under WARP and Walras' law the compensated matrix has a negative-semidefinite symmetric part. Thus a positive Marshallian own-price derivative  $\partial_{p_\ell} x_\ell > 0$  is possible only when the income effect is sufficiently adverse: the good is inferior ( $\partial_w x_\ell < 0$ ), and the income effect dominates the nonpositive compensated effect. This is the Giffen mechanism.

The possibility of a Giffen response does not by itself establish a utility representation. To derive one from data, impose and test the full integrability conditions developed later: budget exhaustion, homogeneity, WARP (or the stronger GARP on finite data), and the relevant regularity assumptions. The compensated response, rather than the Marshallian derivative alone, is the object restricted by revealed preference.

## Chapter 2

# Preference and Choice

### 2.1 From Choice Data to Preference

Choice data, preference relations, and utility representations are three descriptions of the same decision problem. This chapter fixes the logical relations among them before prices and wealth are introduced in the consumer-theory chapter.

### 2.2 Choice environments

Let  $X$  be a nonempty set of alternatives and let  $\mathcal{A} \subseteq 2^X \setminus \{\emptyset\}$  be the family of feasible menus.

**Definition 2.** Choice correspondence. A choice correspondence is a map  $c : \mathcal{A} \rightrightarrows X$  satisfying  $\emptyset \neq c(A) \subseteq A$  for every  $A \in \mathcal{A}$ . A choice structure is the pair  $(\mathcal{A}, c)$ .

Rationalization. A binary relation  $\succeq$  rationalizes  $(\mathcal{A}, c)$  if  $c(A) = \{x \in A : x \succeq y \text{ for every } y \in A\}$  for every  $A \in \mathcal{A}$ . It weakly rationalizes the data if the equality is replaced by inclusion. A rational preference relation is complete and transitive.

### 2.3 Preference relations

**Definition 3.** For a relation  $\succeq \subseteq X \times X$ , define its strict and symmetric parts by  $x \succ y \iff x \succeq y \text{ and not } y \succeq x$  and  $x \sim y \iff x \succeq y \text{ and } y \succeq x$ . If  $\succeq$  is complete and transitive, then  $\succ$  is irreflexive and transitive, while  $\sim$  is an equivalence relation. These elementary implications are the order-theoretic basis for utility representation.

**Proposition 2.** *Finite utility representation. If  $X$  is finite, a preference relation on  $X$  has a*

real-valued utility representation if and only if it is complete and transitive. A representation is unique up to a strictly increasing transformation of its range.

For an infinite commodity or choice space, representation requires additional assumptions such as continuity and a suitable countability condition. The continuous-utility theorem used later is stated in the consumer-theory chapter.

## 2.4 Revealed preference and WARP

**Definition 4.** Direct revealed preference. For a choice structure  $(\mathcal{A}, c)$ , write  $x \succeq^D y$  if there is a menu  $A \in \mathcal{A}$  with  $x \in c(A)$  and  $y \in A$ . Write  $x \succ^D y$  if, in addition,  $y \notin c(A)$ . Let  $\succeq^*$  be the reflexive-transitive closure of  $\succeq^D$ .

Weak Axiom of Revealed Preference (WARP). The data satisfy WARP when  $x \succeq^D y$  implies not  $y \succ^D x$ . Equivalently, if two menus contain both  $x$  and  $y$ , choosing  $x$  from one and strictly choosing  $y$  from the other is ruled out.

**Proposition 3.** *Necessity of WARP. If a complete and transitive preference relation rationalizes a choice structure, then the structure satisfies WARP.*

Proof. If  $x \in c(A)$  and  $y \in A$ , rationalization gives  $x \succeq y$ . If a second menu strictly chose  $y$  over  $x$ , it would give  $y \succ x$ , contradicting the first comparison.

## 2.5 GARP on finite data

**Definition 5.** Generalized Axiom of Revealed Preference (GARP). A finite dataset satisfies GARP when  $x \succeq^* y$  never occurs together with  $y \succ^D x$ . Thus an indirect revealed-preference chain may return to an alternative only when the return comparison is not strict.

**Theorem 1.** *Finite-data rationalizability (cited). For a finite budget dataset with positive prices and observed expenditure exhaustion, GARP is equivalent to weak rationalizability by a complete and transitive preference relation on the observed budgets. Afriat's theorem strengthens this conclusion to a locally nonsatiated, continuous and concave utility representation. The proof is deferred to the Afriat section of the consumer-theory chapter, where the finite inequalities are written explicitly.*

WARP is the one-step restriction; GARP is its transitive closure and is therefore the relevant condition when the data do not contain every pair of alternatives in a common menu. Exact

single-valued rationalization can require additional tie-breaking assumptions, so the theorem above is stated for weak rationalization.

**Example 2.** Let  $X = \{a, b, c\}$  and suppose  $a \succ^D b$ ,  $b \succ^D c$ , and  $c \succ^D a$ . This strict cycle violates GARP and cannot be generated by a transitive preference relation.

The separation between the abstract choice language and the budget-set language is deliberate. In the next chapter, prices and wealth impose additional structure, and the Afriat inequalities provide a finite certificate that can be checked mechanically.

## Chapter 3

# Consumer Theory: Utility Maximization

This chapter derives demand from preference maximization and develops the duality between utility maximization and expenditure minimization. Throughout,  $X = \mathbb{R}_+^L$ ,  $L \geq 1$  is the fixed number of commodities,  $P = \mathbb{R}_{++}^L$ ,  $W = \mathbb{R}_{++}$ .

### 3.1 Demand Correspondences

**Definition 6.** A Walrasian demand correspondence is a map

$$x : P \times W \rightrightarrows X$$

with nonempty values such that every  $q \in x(p, w)$  satisfies  $p \cdot q \leq w$ . If demand is generated by a choice correspondence  $c$ , then  $x = c \circ B$ , where

$$B(p, w) = \{q \in X : p \cdot q \leq w\}.$$

**Example 3.** With two perfect-substitute goods and utility  $u(q) = q_1 + q_2$ ,

$$x(p, w) = \begin{cases} \{(w/p_1, 0)\}, & p_1 < p_2, \\ \{(0, w/p_2)\}, & p_2 < p_1, \\ \{q \in \mathbb{R}_+^2 : p_1 q_1 + p_2 q_2 = w\}, & p_1 = p_2. \end{cases}$$

The equal-price value is the entire budget line, not a pair of independent intervals.

For later use, distinguish the following properties.

- A correspondence has a closed graph if  $(p^n, w^n) \rightarrow (p, w)$ ,  $q^n \in x(p^n, w^n)$ , and  $q^n \rightarrow q$  imply  $q \in x(p, w)$ . Closed graph and upper hemicontinuity are not equivalent without additional compactness or local-boundedness conditions.
- It is compact valued if  $x(p, w)$  is compact for every  $(p, w)$ . In Euclidean space, compactness is equivalent to being closed and bounded.
- It is convex valued if  $\alpha q + (1 - \alpha)r \in x(p, w)$  whenever  $q, r \in x(p, w)$  and  $\alpha \in [0, 1]$ .

### 3.2 Utility Representation

**Definition 7.** A preference relation  $\succeq$  on  $X \subseteq \mathbb{R}^L$  is continuous if its upper and lower contour sets are closed: for every  $y \in X$ ,

$$\{x \in X : x \succeq y\} \quad \text{and} \quad \{x \in X : y \succeq x\}$$

are closed in  $X$ . Equivalently, if  $x^n \rightarrow x$ , then  $x^n \succeq y$  for every  $n$  implies  $x \succeq y$ , and  $y \succeq x^n$  for every  $n$  implies  $y \succeq x$ .

**Lemma 1.** *Every total preorder on a countable set has a real-valued utility representation.*

*Proof.* Let  $I$  be the symmetric part of the preorder. The quotient  $X/I$  is countable, and the relation induced on it is a linear order by Appendix A. Every countable linear order admits an order embedding into  $\mathbb{Q}$ : enumerate its elements and insert the image of each new element at a rational number between the images of its already embedded lower and upper neighbors. Composing this embedding with the quotient map gives  $u : X \rightarrow \mathbb{Q} \subseteq \mathbb{R}$  such that  $x \succeq y \iff u(x) \geq u(y)$ .  $\square$

**Theorem 2.** *(Continuous utility representation) If  $\succeq$  is a continuous total preorder on  $X = \mathbb{R}_+^L$ , then there exists a continuous function  $u : X \rightarrow \mathbb{R}$  such that, for every  $x, y \in X$ ,*

$$x \succeq y \iff u(x) \geq u(y).$$

*Proof.* This is Debreu's continuous-utility theorem specialized to Euclidean commodity space. The substantive topological input is that  $\mathbb{R}_+^L$  is second countable and connected. Continuity of the preorder and the gap lemma provide a countable order-dense family of indifference classes.

Mapping those classes into a countable dense subset of  $\mathbb{R}$  and extending across the gaps yields an order-preserving continuous map. The result cannot be obtained by taking arbitrary limits of a utility defined only on  $\mathbb{Q}^L$ : such limits need not exist or be independent of the approximating sequence.  $\square$

The theorem guarantees existence, not uniqueness. If  $u$  represents  $\succeq$  and  $f$  is strictly increasing on the range of  $u$ , then  $f \circ u$  represents the same preference relation.

### 3.3 The Utility Maximization Problem

We assume that  $u$  is continuous and  $X = \mathbb{R}_+^L$ .

The utility maximization problem (UMP) is:

$$\max_{x \geq 0} u(x)$$

$$s.t. \quad p \cdot x \leq w.$$

Alternatively we can define the restriction as  $x \in B(p, w) = \{x \in \mathbb{R}^L : p \cdot x \leq w\}$ .

**Proposition 4.** *If  $p \gg 0$  and  $u$  is continuous, then the UMP has a solution.*

*Proof.* Check  $B(p, w)$  is compact, in this case it is equivalent to proving that it is closed and bounded.

It is bounded because  $x_l \leq w/p_l$  for all  $l$ . It is closed because  $x^n$ , such that  $p \cdot x^n \leq w$  and if  $x^n \rightarrow x$  it must be the case that  $\lim_{n \rightarrow \infty} p \cdot x^n \leq \lim_{n \rightarrow \infty} w$  implies that  $p \cdot x \leq w$ .

The Weierstrass maximum theorem therefore gives a solution to the UMP.  $\square$

We now prove further properties of the demand correspondence.

**Proposition 5.** *The Marshallian demand correspondence  $x(p, w) \equiv \arg \max_y \{u(y) | y \in B(p, w)\}$  is non-empty, compact valued and upper hemi-continuous. Moreover,  $v(p, w) = u(x)$  for  $x \in x(p, w)$  (i.e., the indirect utility) is continuous.*

*Proof.* The result follows from Berge's maximum theorem once we verify that the correspondence  $B(\cdot)$  is a continuous correspondence. We already know that  $u$  is continuous and that  $B(p, w)$  is a compact set, thus making the  $B(\cdot)$  correspondence compact valued.

Take  $z^n = (p^n, w^n)$  now assume that  $z^n \rightarrow z$ , now we want to prove that  $B(z^n) \rightarrow B(z)$  where convergence is understood in the usual set-convergence sense. Recall that,  $B(p^n, w^n) = \{x \in X | p^n \cdot x \leq w^n\}$ . First assume that  $x^n \in B(z^n)$  and  $x^n \rightarrow x$ . We prove that  $x \in B(z)$ . If  $x^n \in B(z^n)$  then  $p^n \cdot x^n \leq w^n$  for all  $n$ , applying the limits over this inequality we conclude that  $p \cdot x \leq w$  where  $x^n \rightarrow x$  and  $(p^n, w^n) \rightarrow (p, w)$ . Then  $x \in B(z)$  with  $z = (p, w)$ .

For the reverse inclusion, let  $x \in B(p, w)$ . If  $p \cdot x < w$ , then  $x \in B(p^n, w^n)$  for all sufficiently large  $n$ . If  $p \cdot x = w$ , choose  $x^n = (1 - \varepsilon_n)x$  with  $\varepsilon_n \downarrow 0$  slowly enough that  $p^n \cdot x^n \leq w^n$ . Hence the budget correspondence is continuous on  $\mathbb{R}_{++}^L \times \mathbb{R}_{++}$ , and Berge's theorem gives the stated properties. □

We need some definitions.

**Definition 8.** Local nonsatiation. A preference relation  $\succeq$  is locally nonsatiated if, for every  $x \in X$  and every  $\varepsilon > 0$ , there is  $y \in X$  with  $\|y - x\| < \varepsilon$  and  $y \succ x$ .

**Proposition 6.** If a locally nonsatiated utility function generates the Marshallian demand correspondence, then every demanded bundle exhausts the budget:  $p \cdot x = w$  for every  $x \in x(p, w)$ .

Proof. If  $p \cdot x < w$ , local nonsatiation supplies a strictly preferred bundle arbitrarily close to  $x$ . For a sufficiently small neighborhood the bundle remains affordable, contradicting utility maximization. Therefore the budget constraint binds.

**Proposition 7.** The same demand correspondence is homogeneous of degree zero. If preferences are convex, demand is convex-valued; if preferences are strictly convex, demand is single-valued.

### 3.4 KKT Conditions for the UMP

Under differentiability, the first-order conditions characterize an interior solution.

We write the KKT **FOC** necessary conditions.

$$\mathcal{L} = u(x) + \lambda(w - p \cdot x)$$

$$\partial_{x_l} u(x^*) \leq \lambda^* p_l \text{ and } x_l^* [\partial_{x_l} u(x^*) - \lambda^* p_l] = 0 \text{ for all } l \in \{1, \dots, L\}.$$

$$w - p \cdot x^* \geq 0 \text{ and } \lambda^*(w - p \cdot x^*) = 0.$$

This is a system with  $L + 1$  unknowns.

Local nonsatiation implies that the consumer is on the budget hyperplane (the frontier). This means that

$$w - p \cdot x^* = 0.$$

For an interior solution (all quantities positive), the KKT conditions reduce to

$$\frac{\partial_{x_l} u(x^*)}{\partial_{x_k} u(x^*)} = \frac{p_l}{p_k} \quad \forall l, k \in \{1, \dots, L\}.$$

Note that for  $L = 2$ ,  $-\frac{p_1}{p_2}$  is the slope of the budget line and  $-\frac{\partial_{x_1} u(x^*)}{\partial_{x_2} u(x^*)}$  is the slope of the indifference curve, usually this is also referred as the MRS.

### 3.5 Properties of the Indirect Utility Function

The indirect utility is the maximal utility achieved given the budget constraint  $v(p, w) = \max_{x \geq 0} u(x)$  s.t.  $p \cdot x \leq w$ .  $v : P \times W \rightarrow \mathbb{R}$  is a form of ranking prices and income (budgets).

Alternatively it is the utility evaluated in the Marshallian demand  $v(p, w) = u(x(p, w))$ .

**Proposition 8.** *Suppose that  $u(\cdot)$  is a continuous utility function representing a locally non-satiated preference relation  $\succeq$  defined on the consumption set  $X = \mathbb{R}_+^L$ . The indirect utility function  $v(p, w)$  is:*

1. *Homogeneity of degree zero.*
2.  *$v(p, w)$  is strictly increasing in  $w$  and non-increasing in  $p_l$ .*
3.  *$v(p, w)$  is quasiconvex in  $(p, w)$ . The set  $\{(p, w) | v(p, w) \leq \bar{v}\}$  is convex.*
4.  *$v(p, w)$  is continuous in  $p$  and  $w$ .*

Proof. Existence, homogeneity, monotonicity, and continuity follow from the compact budget correspondence and Berge's maximum theorem.

For quasiconvexity, fix  $\bar{v}$  and let  $L_{\bar{v}} = \{(p, w) : v(p, w) \leq \bar{v}\}$ . If  $u(x) > \bar{v}$ , then every  $(p, w) \in L_{\bar{v}}$  must satisfy  $p \cdot x > w$ , because otherwise  $x$  would be affordable and the value would exceed  $\bar{v}$ . Conversely, this strict inequality for every  $x$  with  $u(x) > \bar{v}$  rules out an affordable bundle with utility above  $\bar{v}$ . Hence

$$L_{\bar{v}} = \bigcap_{u(x) > \bar{v}} \{(p, w) : p \cdot x > w\}.$$

Each set in the intersection is a linear half-space in  $(p, w)$ , so the lower contour set is convex and  $v$  is quasiconvex.

### 3.6 Roy's Identity

Recall the definition of indirect utility  $v(p, w) = u(x(p, w))$ .

Assume the maximizer is unique and the value function is differentiable. The envelope theorem applied to

$$v(p, w) = \max_{x \geq 0, p \cdot x \leq w} u(x)$$

gives

$$\partial_w v(p, w) = \lambda(p, w), \quad \nabla_p v(p, w) = -\lambda(p, w)x(p, w).$$

This provides the Roy's identity (this result only works under differentiability and uniqueness)

$$x(p, w) = -\frac{\nabla_p v(p, w)}{\partial_w v(p, w)}.$$

**Example 4.** Quasilinear demand and Roy's identity. Let good  $L$  be the numeraire, set  $p_L = 1$ , and let utility be  $u(q, q_L) = g(q) + q_L$  for  $q \in \mathbb{R}_+^{L-1}$ . The indirect utility is

$$v(p, w) = w + \max_{q \geq 0} \{g(q) - p_{-L} \cdot q\}.$$

When the maximizer is unique and the envelope theorem applies, Roy's identity gives  $x_k(p, w) = -\partial_{p_k} v(p, w)$  for  $k < L$ , and the numeraire demand is the residual  $x_L = w - p_{-L} \cdot x_{-L}(p)$ . This example has no income effect on the non-numeraire demands.

### 3.7 Expenditure Minimization Problem (EMP)

The Expenditure Minimization Problem (EMP) asks the question, if prices are  $p$ , what is the minimum amount the consumer would have to spend to achieve utility level  $\bar{u}$ . That is:

$$\min_{x \geq 0} p \cdot x$$

s.t.

$$u(x) \geq \bar{u}.$$

**Proposition 9.** Suppose  $u$  is a continuous utility function representing a locally nonsatiated preference relation  $\succeq$  defined on the consumption set  $X = \mathbb{R}_+^L$  and the price vector is  $p \gg 0$ . We

have.

(i) If  $x^*$  is optimal in the UMP when  $w > 0$  then  $x^*$  is optimal in the EMP when the required level of utility is  $u(x^*)$ . Moreover, the minimized expenditure level  $p \cdot x^* = w$  is exactly the wealth  $w$ .

(ii) If  $x^*$  is optimal in the EMP when the required level of utility is  $u > u(0)$  then  $x^*$  is optimal in the UMP when wealth is  $p \cdot x^*$ . Moreover, the maximized utility level  $u(x^*) = u$  is exactly  $u$ .

We omit the proof. For a complete proof, see the Afriat treatment in the references.

**Expenditure function** ( $e : P \times \mathbb{R} \rightarrow \mathbb{R}$ ) is  $e(p, u) = p \cdot x^*$  where  $x^*$  is an element in the solution to the EMP.

**Proposition 10.** Suppose that  $u$  is a continuous utility function representing a locally nonsatiated preference relation  $\succeq$  defined on the consumption set  $X = \mathbb{R}_+^L$ , with  $p \in \mathbb{R}_{++}^L$  and the target utility in the attainable range. The expenditure  $e(p, u)$  is

(i) Homogeneous of degree one in  $p$ .

$$\min_{x \geq 0} \alpha p \cdot x \text{ s.t. } u(x) \geq \bar{u}.$$

(ii) Strictly increasing in  $u$  and non-decreasing in  $p_l$  for all  $l$

(iii) Concave in  $p$

(iv) Continuous in  $p, u$ .

*Proof.* (i),(ii) are omitted.

(iii) Concavity.

We want to show that  $e(p, u)$  is concave in  $p$  for any  $u$ .

$$e(p^\alpha, \bar{u}) \geq \alpha e(p, \bar{u}) + (1 - \alpha)e(p', \bar{u}),$$

when  $p^\alpha = \alpha p + (1 - \alpha)p'$  for some  $\alpha \in [0, 1]$ .

Let  $h(p, \bar{u})$  be the minimizer of the EMP, we abuse notation and let it denote any element of the set of optimizers at the given prices:

$$\begin{aligned} e(p^\alpha, \bar{u}) &= p^\alpha \cdot h(p^\alpha, \bar{u}) \\ &= \alpha p \cdot h(p^\alpha, \bar{u}) + (1 - \alpha)(p') \cdot h(p^\alpha, \bar{u}) \end{aligned}$$

Note that  $e(p, \bar{u}) = p \cdot h(p, \bar{u}) \leq p \cdot h(p^\alpha, \bar{u})$ , and we can conclude that:

$$e(p^\alpha, \bar{u}) \geq \alpha p \cdot h(p, \bar{u}) + (1 - \alpha)(p') \cdot h(p', \bar{u}).$$

(iv) Continuity in  $p, u$ .

Continuity follows from the standard expenditure-function theorem. The unbounded feasible set cannot be inserted directly into Berge's theorem. Instead, around any  $(p, \bar{u})$  with strictly

positive prices, one first obtains a common expenditure upper bound from one feasible bundle. Strict positivity of nearby prices then confines every optimizer to a common compact box. Berge's theorem applies to this locally compactified problem. Continuity of preferences and local nonsatiation ensure feasibility stability and that an optimum reaches the target utility.

□

### 3.8 Hicksian Compensated Demand

The Hicksian Compensated Demand or Hicksian demand is  $h(p, \bar{u}) = \arg \min_{x \geq 0, u(x) \geq \bar{u}} p \cdot x$ .

**Proposition 11.** *Suppose  $u(\cdot)$  is continuous and it represents a locally nonsatiated preference relation  $\succeq$ . Then the Hicksian demand correspondence  $h(p, u)$  has the following properties:*

- (i) *HD0 in  $p$*
- (ii) *No excess utility: For  $x^* \in h(p, u)$ ,  $u(x^*) = \bar{u}$*
- (iii) *Convexity/Uniqueness: If  $\succeq$  is convex, then  $h(p, u)$  is a convex set, and if  $\succeq$  is strictly convex, then  $h(p, u)$  is a unique element.*

### 3.9 KKT Conditions for the EMP

Assume  $u$  is twice continuously differentiable.

The Lagrangian for this problem is:

$$\mathcal{L}_{EMP} = p \cdot x - \lambda(u(x) - \bar{u}).$$

The FOC are given by

$$p \geq \lambda \nabla_x u(x^*)$$

$$x_l^* [p_l - \lambda \nabla u(x^*)_l] = 0,$$

for some  $\lambda \geq 0$ .

### 3.10 Sheppard's Lemma

**Proposition 12.** *Every Hicksian bundle is a supergradient of the concave expenditure function in prices. If  $e(\cdot, \bar{u})$  is differentiable at  $p$ , Hicksian demand is unique there and Shepard's identity holds:*

$$\nabla_p e(p, \bar{u}) = h(p, \bar{u}).$$

*Proof.* At a differentiability point, apply the envelope theorem to the constrained problem.

$$e(p, \bar{u}) = p \cdot h + \lambda^*(u(h) - \bar{u})$$

We derive with respect to  $p$  and we obtain:

$$\nabla_p e(p, \bar{u}) = h(p, \bar{u})$$

□

### 3.11 Additional Properties

Assume throughout this subsection that  $e(\cdot, u)$  is twice continuously differentiable, so Hicksian demand is single-valued and differentiable. Let  $D_p h(p, u)$  (Jacobian)  $D_p h(p, u)_{k,l} = \partial_{p_l} h_k(p, u)$  be the  $L \times L$  matrix of derivatives of prices.

**Proposition 13.** *The Hicksian demand function has the following properties:*

1.  $D_p h(p, u) = \mathcal{H}e(p, u)$  ( $\mathcal{H}e(p, u)_{k,l} = \partial_{p_k} \partial_{p_l} e(p, u)$ ).

The derivative with respect to prices of the Hicksian demand is the Hessian of the expenditure function.

2.  $D_p h(p, u)$ .

This is a NSD matrix. This follows from the fact that a Hessian of a concave function is NSD.

3.  $D_p h(p, u)$  is symmetric.

Young's theorem (continuity of the second derivatives of  $e$ ,  $\partial_{p_k} \partial_{p_l} e(p, u) = \partial_{p_l} \partial_{p_k} e(p, u)$ ).

4.  $D_p h(p, u)p = 0$ .

This follows from HD0 of  $h(p, \bar{u})$  in  $p$  and Euler's identity.

**Definition 9.** NSD (negative semidefinite matrix) For any vector  $v \in \mathbb{R}^L$ , a symmetric matrix  $A$  of size  $L \times L$  is NSD if and only if  $v^\top A v \leq 0$ .

**Example 5.** The following matrix is NSD:

$$A = \begin{bmatrix} -2 & 0 \\ 0 & 0 \end{bmatrix}$$

$$v'Av = -2v_1^2 \leq 0$$

for any  $v \in \mathbb{R}^2$ .

### 3.12 Summary of Relationships between the EMP and UMP

The **expenditure function** is the minimum  $e(p, \bar{u}) = p \cdot h(p, \bar{u})$  value attained at the given prices and target utility.

There is a **duality relation** between the EMP and the UMP that we will not explore at this point. However, we will state the following relations.

$$h(p, v(p, w)) = x(p, w)$$

$$x(p, e(p, u)) = h(p, u).$$

Also,

$$\bar{u} = v(p, e(p, \bar{u}))$$

$$w = e(p, v(p, w)).$$

### 3.13 Slutsky Equation

The Slutsky matrix is  $S : P \times W \rightarrow \mathbb{R}^{L \times L}$ ,  $S(p, w)_{l,k} = \partial_{p_k} x_l(p, w) + \partial_w x_l(p, w) x_k(p, w)$ ,  $S(p, w) = D_p x(p, w) + D_w x(p, w) x(p, w)^\top$ .

Consider price compensation that keeps the original bundle  $q$ :  $D_p[x(p, p \cdot q)] = D_p x(p, p \cdot q) + D_w x(p, p \cdot q) q^\top$ . Let  $q = x(p, w)$  for the original price and income and then replace to obtain the Slutsky matrix (under Walras' law):

$$D_p x(p, p \cdot q)|_{q=x(p,w)} = S(p, w).$$

We establish now a very useful equation.

$$h(p, u) = x(p, e(p, u))$$

Differentiating with respect to  $p$

$$D_p h(p, u) = D_p x(p, e(p, u)) + D_w x(p, e(p, u)) \nabla_p e(p, u)^\top$$

By Sheppard's lemma and by the identities that relate the expenditure and wealth

$$D_p h(p, v(p, w)) = D_p x(p, w) + D_w x(p, w) x(p, w)^\top = S(p, w).$$

### 3.14 Summary of the Implications of Utility Maximization

**Proposition 14.** *If a demand function  $x(p, w)$  is generated by maximizing a complete, transitive, continuous and locally nonsatiated preference relation represented by  $u$  subject to a linear budget constraint (that is,  $x(p, w) \in \arg \max_{y \in B(p, w)} u(y)$  for every  $(p, w) \in P \times W$  for some  $u : X \rightarrow \mathbb{R}$ ). Then: (i)  $x(p, w)$  satisfies Walras' law, (ii)  $x(p, w)$  is homogeneous of degree zero.*

*If, in addition,  $x$  is continuously differentiable and the expenditure function is twice differentiable so that the Slutsky matrix is defined, its Slutsky matrix function  $S(p, w)$  has a negative-semidefinite symmetric part: (i)  $a^\top S(p, w) a \leq 0$  for every  $a \in \mathbb{R}^L$ . (ii) Singular in prices  $S(p, w)p = 0$ , (iii) Symmetric  $S(p, w) = S(p, w)^\top$  for all  $p, w \in P \times W$ .*

This proposition summarizes our previous results.

### 3.15 Integrability

We now ask for sufficient conditions under which an observed demand function is generated by utility maximization. In the abstract choice setting, WARP and GARP restrict rationalizability. Competitive budget sets are a smaller domain than arbitrary menus, so differentiable demand requires additional restrictions. The relevant restrictions are stated in terms of the Slutsky matrix.

Of course, we assume in this subsection that the demand function  $x$  is continuously differentiable.

Recall that the Slutsky matrix function is  $S(p, w) = D_p x(p, w) + D_w x(p, w)x(p, w)^\top$ , or in the scalar version  $s_{ij}(p, w) = \partial_{p_j} x_i(p, w) + \partial_w x_i(p, w)x_j(p, w)$  for all  $i, j \in \{1, \dots, L\}$ .

**Theorem 3.** (*Local integrability theorem, cited.*) *Let  $D \subseteq \mathbb{R}_{++}^L \times \mathbb{R}_{++}$  be an open, simply connected domain, and let  $x : D \rightarrow \mathbb{R}_+^L$  be continuously differentiable. Suppose that  $x$  satisfies Walras' law and homogeneity of degree zero, and that its Slutsky matrix is symmetric and negative semidefinite on  $D$ . Under the standard nondegeneracy and boundary conditions ensuring that the associated differential system has a unique solution through each point, these conditions are locally sufficient for a locally nonsatiated utility representation. A global representation requires the corresponding global coverage and boundary conditions.*

The qualifications are essential. Symmetry and negative semidefiniteness alone, stated on an unspecified domain, do not guarantee a global representation: one must control the domain, boundary behavior, and the differential equation's existence and uniqueness. Conversely, every continuously differentiable demand generated by a regular locally nonsatiated utility satisfies Walras' law, homogeneity, symmetry, and negative semidefiniteness of the Slutsky matrix.

*Remark.* Some textbooks require a boundedness condition on the elasticity of wealth, this is in general not required.

### 3.16 Examples and Applications of Demand Integrability

There are two commodities  $L = 2$  and you estimate a demand function  $x_1(p, w) = \frac{\alpha w}{p_1}$ , and you know that the demand of good two is given as a residual. Furthermore, we have checked that this demand has a Slutsky matrix that is symmetric, NSD and singular in prices, we normalize  $p_2 = 1$ . Thus  $x_2 = (1 - \alpha)w$ .

The  $L - 1$  differential equation is:

$$\mu'(p_1, 1) = \frac{\alpha \mu(p_1, 1)}{p_1} = x_1((p_1, 1), \mu((p_1, 1))),$$

for short, we can write:

$$\mu'(p_1) = \frac{\alpha \mu(p_1)}{p_1} = x_1(p_1, \mu(p_1)),$$

We can write more simply:

$$\frac{\mu'}{\mu} = \frac{\alpha}{p_1}.$$

Simple integration provides a family of solutions:

$$\ln \mu = \alpha \ln p_1 + C,$$

and in levels:

$$\mu = K p_1^\alpha,$$

with  $K = \exp(C)$  a constant of integration.

Given some initial conditions  $(p_1^0, w^0)$ , we must have:

$$w^0 = K (p_1^0)^\alpha, \quad K = \frac{w^0}{(p_1^0)^\alpha},$$

or

$$\mu(p_1^*; p_1^0, w^0) = \frac{w^0}{(p_1^0)^\alpha} (p_1^*)^\alpha.$$

Now let's define the indirect utility, with  $p_1^* = 1$ :

$$v(p_1, w) = \mu(p^*, p_1, w) = \frac{w}{p_1^\alpha}.$$

To recover the utility  $u$ , we need to invert the demand, note that our primitive is indeed invertible.

We have to use the demand function and obtain the inverse demand:

$$x_1(p, w) = \frac{\alpha w}{p_1}; \quad x_2(p, w) = (1 - \alpha)w$$

$$\frac{x_1}{x_2} = \frac{\alpha}{(1 - \alpha)} \frac{1}{p_1}$$

$$p_1 = \frac{\alpha}{1 - \alpha} \frac{x_2}{x_1}, \quad w = \frac{x_2}{1 - \alpha}.$$

Replacing back in the indirect utility we have:

$$u(x_1, x_2) = v \circ (p_1, w)(x_1, x_2) = v\left(\frac{\alpha}{1-\alpha} \frac{x_2}{x_1}, \frac{x_2}{1-\alpha}\right).$$

$$u(x_1, x_2) = \frac{\frac{x_2}{1-\alpha}}{\left(\frac{\alpha}{1-\alpha} \frac{x_2}{x_1}\right)^\alpha} = \left(\frac{x_2}{1-\alpha}\right)^{1-\alpha} \left(\frac{x_1}{\alpha}\right)^\alpha$$

$$u(x_1, x_2) = \hat{c} x_1^\alpha x_2^{1-\alpha}, \quad \hat{c} = (1-\alpha)^\alpha \alpha^\alpha.$$

This is a Cobb-Douglas utility function.

### 3.17 Generalized Axiom of Revealed Preference (GARP)

We now move from unrestricted demand functions to finite observations. Suppose that the dataset is  $\mathcal{O} = \{p^k, x^k\}_{k=1}^K$ ,  $K \geq 2$ . Walras' law is maintained throughout this section.

**Definition 10.** We say that  $x^t$  is directly revealed preferred to  $x$  ( $x^t R^D x$ ) if and only if  $p^t \cdot x^t \geq p^t \cdot x$ .

Furthermore, we define an (indirect) notion of revealed preference.

**Definition 11.** We say that  $x^t$  is (indirectly) revealed preferred to  $x$  ( $x^t R x$ ) if and only if there is a sequence of commodities such that  $x^t R^D x^1 R^D x^2 R^D \dots x^n R^D x$ .

Using some ideas from order theory we can see that  $R$  is the transitive closure of  $R^D$ .

In the previous treatment of rationality we did not allow for correspondences; we only allowed for functions. But what happens for cases when we observe for the same prices  $p^t = p^s$  different demand choices  $x^t \neq x^s$  such that  $p^t \cdot x^t = p^s \cdot x^s$ , recall that this is a case where from the same budget set two different choice bundles are chosen (this can be seen as the case of a correspondence generating the data).

**Definition 12.** We say that a utility function  $u$  (weakly) rationalizes a dataset  $\mathcal{O}$ , whenever for all  $t \in \{1, \dots, K\}$ ,  $u(x^t) \geq u(x)$  for all  $p^t \cdot x^t \geq p^t \cdot x$ .

We now introduce a property that serves as a test of rationalizability. We can define the strict direct preference relation as  $x^t P^D x^s$  if and only if  $p^t \cdot x^t > p^t \cdot x^s$

**Definition 13.** A finite demand dataset  $\mathcal{O} = \{(p^t, x^t)\}_{t=1}^K$  satisfies the generalized axiom of revealed preference (GARP) if

$$x^t R x^s \implies \neg(x^s P^D x^t)$$

for all observations  $s, t$ , where  $R$  is the transitive closure of direct revealed preference and

$$x^s P^D x^t \iff p^s \cdot x^s > p^s \cdot x^t.$$

Equivalently, a chain of weak revealed-preference comparisons may not be closed by a strict direct comparison in the opposite direction.

**Proposition 15.** (*Afriat's Theorem*) Given  $\mathcal{O} = \{p^k, x^k\}_{k=1}^K$ , the following are equivalent:

1. There exists a locally nonsatiated utility function  $u$  that rationalizes the data in the sense that for all  $k$ ,  $u(x^k) \geq u(x)$  for all  $x$  such that  $p^k \cdot x^k \geq p^k \cdot x$ .
2. The dataset  $\mathcal{O} = \{p^k, x^k\}_{k=1}^K$  satisfies GARP.
3. There exist numbers  $u^t \in \mathbb{R}$  and  $\lambda^t > 0$  to the set of linear inequalities

$$u^t \leq u^s + \lambda^s p^s \cdot (x^t - x^s) \forall s, t \in \{1, \dots, K\}$$

4. There exists a continuous, concave, and strictly monotonic, utility function  $u$  that rationalizes the data in the sense that for all  $k$ ,  $u(x^k) \geq u(x)$  for all  $x$  such that  $p^k \cdot x^k \geq p^k \cdot x$ .

For a differentiable rationalization, (4) implies (3) through supporting first-order inequalities; at a boundary coordinate, the KKT inequality is multiplied by a nonnegative observed quantity. For a nonsmooth concave rationalization, the same argument uses a positive supergradient at each observation.

If  $u$  is concave then  $u(x^t) \leq u(x^s) + \nabla u(x^s) \cdot (x^t - x^s)$ . Given that  $u$  is assumed smooth  $\nabla u$  exists (super-gradient for concave functions). Now, by the KKT conditions for the UMP,  $\nabla u(x^s) \leq \lambda^s p^s$ . This implies that

$$u(x^t) \leq u(x^s) + \lambda^s p^s \cdot (x^t - x^s).$$

$$u(x^t) = u^t.$$

*Proof.* (1)  $\implies$  (2), (2)  $\implies$  (3), (3)  $\implies$  (4), (4)  $\implies$  (1).

(1) implies (2),

Assume that  $\mathcal{O}$  is rationalized by a nonsatiated utility function and that GARP fails, then  $x^1 R x^2 \dots R x^n$  and  $x^n P^D x^1$  with strict preference for the last one, this means that  $u(x^1) \geq u(x^2) \geq \dots \geq u(x^n)$  and also that  $p^n \cdot x^n > p^n \cdot x^1$  but this means that there is a  $\|y - x^1\| < \epsilon$  for some  $\epsilon > 0$  such that  $p^n \cdot x^n > p^n \cdot y$  and such that  $u(y) > u(x^1)$  by local nonsatiation. This implies that  $u(y) > u(x^n)$  which means that  $x^n$  cannot be a maximizer for budget  $B(p^n, p^n \cdot x^n)$ , this is a contradiction.

(2) implies (3)

The implication from GARP to the Afriat inequalities is the finite Afriat feasibility lemma. Write the inequalities as a finite linear feasibility problem in  $(u^1, \dots, u^K, \lambda^1, \dots, \lambda^K)$ , with  $\lambda^t > 0$ . Farkas' lemma says that failure of feasibility is equivalent to a nonnegative dual certificate. After collecting the price-expenditure terms along the directed revealed-preference graph, such a certificate is exactly a revealed-preference cycle with at least one strict reversal. GARP excludes every such cycle, so the system is feasible. Conversely, summing the Afriat inequalities around a revealed-preference cycle rules out a strict reversal. This gives the equivalence between statements (2) and (3).

(3) implies (4)

Let  $u(x) = \min_{t \in \{1, \dots, K\}} \{u^t + \lambda^t p^t \cdot (x - x^t)\}$ , this function is continuous in  $u$  and concave, it is also strictly increasing in  $x$  (given that  $\lambda^t > 0$ , and  $p^t \gg 0$ ).

To verify rationalization, first observe that  $u(x^t) = u^t$  for every observation  $t$ . Indeed, the Afriat inequality gives

$$u^t \leq u^m + \lambda^m p^m \cdot (x^t - x^m)$$

for every  $m$ , while the term with  $m = t$  equals  $u^t$ . Therefore the minimum of these affine terms is exactly  $u^t$ .

It remains to show that  $u(x) \leq u(x^s)$ , if  $p^s \cdot x^s \geq p^s \cdot x$

If  $p^s \cdot x^s \geq p^s \cdot x$ , or  $p^s \cdot (x^s - x) \geq 0$ , then  $u(x) = \min_t \{u^t + \lambda^t p^t \cdot (x - x^t)\} \leq u^s + \lambda^s p^s \cdot (x - x^s) \leq u^s = u(x^s)$ .

Thus  $u(x^s) \geq u(x)$  when  $p^s \cdot x^s \geq p^s \cdot x$ .

(4) implies (1).

If  $u$  is strictly monotonic then it is locally nonsatiated.

□

### 3.18 Revealed Preference and the Strong Axiom

The preceding section used GARP to characterize weak rationalization, allowing ties in observed choice. This section records the corresponding strict result: with single-valued choice and strict concavity, SARP replaces GARP and the Afriat inequalities become strict. The two results are parallel but answer different rationalizability questions.

**Definition 14.** (Strong Axiom of Revealed Preference -SARP-) The revealed preference relation  $R$  is acyclic, i.e., If  $x^t R x^s$  then it is not the case that  $x^s R^D x^t$  ( $t \neq s$ ).

This is a result by Afriat/Varian/Houtaker.

**Proposition 16.** (*Afriat's Theorem*) Given  $\mathcal{O} = \{p^k, x^k\}_{k=1}^K$ , the following are equivalent:

1. There exists a continuous, nonsatiated utility function  $u$ , strictly concave that rationalizes the data in the sense that for all  $k$ ,  $u(x^k) > u(x)$  for all  $x \neq x^k$  such that  $p^k \cdot x^k \geq p^k \cdot x$ .
2. The dataset  $\mathcal{O} = \{p^k, x^k\}_{k=1}^K$  satisfies SARP.
3. There exist numbers  $u^t \in \mathbb{R}$  and  $\lambda^t > 0$  to the following system of linear inequalities:

$$u^t < u^s + \lambda^s p^s \cdot (x^t - x^s) \forall s, t \in \{1, \dots, K\}, s \neq t$$

### 3.19 Forecasting with Varian's Support Set

**Definition 15.** Given observed data  $\mathcal{O} = \{(p^t, x^t)\}_{t=1}^K$  and a new price-wealth pair  $(p^{K+1}, w^{K+1})$ , Varian's support set is

$$S(p^{K+1}, w^{K+1}) = \left\{ x \in X : \begin{array}{l} \mathcal{O} \cup \{(p^{K+1}, x)\} \text{ satisfies GARP,} \\ p^{K+1} \cdot x = w^{K+1} \end{array} \right\}.$$

The definition already gives the exact criterion: a candidate bundle belongs to the support set if it exhausts the new budget and the augmented dataset satisfies GARP. The transitive closure must be checked, not only one-step comparisons. Whenever an old bundle  $x^t$  is indirectly revealed preferred to the candidate, GARP requires  $p^{K+1} \cdot x \leq p^{K+1} \cdot x^t$ .

### 3.20 Upper Bounds for Welfare Analysis

**Definition 16.** A money metric utility function is defined as  $m(p, x) = e(p, u(x))$  where  $e$  is the expenditure function associated with  $u$ ,  $e(p, \bar{u}) = \min_{z \in X} \{p \cdot z : u(z) \geq \bar{u}\}$ .

For any given  $p$ ,  $m(p, x)$  is a utility function that represents the same preferences as  $u$  but is expressed in money units.

Given a finite dataset  $\mathcal{O}$  Varian proposes the following upper bound for money-metric utility:

$$m^+(p, x) = \min_t p x^t, \quad \text{s.t. } x^t R x.$$

### 3.21 Experiments about Testing Rationality

(This material is presented in Quah<sup>1</sup>)

- Data from the portfolio choice experiment in Choi, Fisman, Gale, and Kariv (AER, 2007).
- 93 undergraduate subjects participated in the experiment at UC Berkeley, each completing 50 decision problems under risk.
- There were two states of the world, each occurring with a known probability, and two Arrow-Debreu securities, one for each state.
- In each decision problem, every subject was given a budget; income was normalized to one and state prices were chosen at random. 47 subjects were subjected to the symmetric treatment, where  $\pi_1 = \pi_2 = \frac{1}{2}$ . The rest had the asymmetric treatment where  $\pi_1 = \frac{1}{3}$ ,  $\pi_2 = \frac{2}{3}$  (or vice versa).

### 3.22 Results

- In the symmetric treatment, 12/47 pass GARP.
- In the asymmetric treatment, 4/46 pass GARP.
- Roughly 17% pass GARP.

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<sup>1</sup><http://www.johnquah.com/uploads/7/5/6/7/75676677/survey-qutweb.pdf>

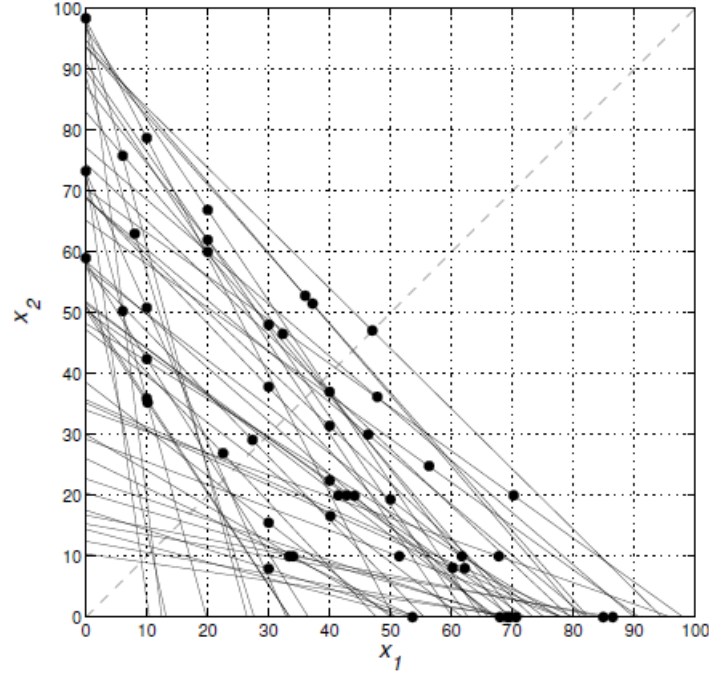


Figure: Subject 216 (symmetric treatment)

Figure 3.1: Subject 216 (symmetric treatment). Source: Quah.

### 3.23 Afriat's Cost-Efficiency Index

Suppose a dataset  $O$  cannot be rationalized by any  $u$  (GARP/SARP fails).

Now consider the case where we shrink budget sets implied by  $O$  by an efficiency index  $\alpha \in [0, 1]$ :

Let me define  $\alpha \in [0, 1]$ :

$$\alpha p^t \cdot x^t \geq p^t \cdot x^s \iff x^t R_\alpha^D x^s$$

I have a  $\mathcal{O}$  that fails rationality and in particular I observe that  $x^1 R^D x^2 P^D x^1$ ,  $\mathcal{O}/\{p^1, x^1\}$  this satisfies GARP.

Notice that every dataset  $\mathcal{O}$  will satisfy GARP( $R_\alpha$ ) for some  $\alpha \in [0, 1]$ , find the  $\alpha^* = \max\{\alpha : \mathcal{O} \text{ satisfies GARP on } R_\alpha\}$ .

Then every dataset  $O$  can be  $\alpha$ -rationalized if we find  $u(x^t) \geq u(x)$  for all  $x \in B^t(\alpha)$ , where  $B^t(\alpha) = \{x \in \mathbb{R}_+^L : p^t \cdot x \leq \alpha p^t \cdot x^t\}$ .

**Definition 17.** The Critical Cost Efficiency Index (CCEI) of a dataset  $O$  is  $\alpha^* = \sup\{\alpha \in [0, 1] : \alpha \in A\}$  where  $\alpha \in A$  if there is a  $u$  that rationalizes the data  $O(\alpha) = \{B^t(\alpha), x^t\}_{t=0}^T$ . It amounts to checking SARP/GARP for a grid of  $\alpha$  values and checking whether it rationalizes the data.

The supremum of those  $\alpha$  numbers is the CCEI.

Now, The CCEI distribution for the experiment is shown below.

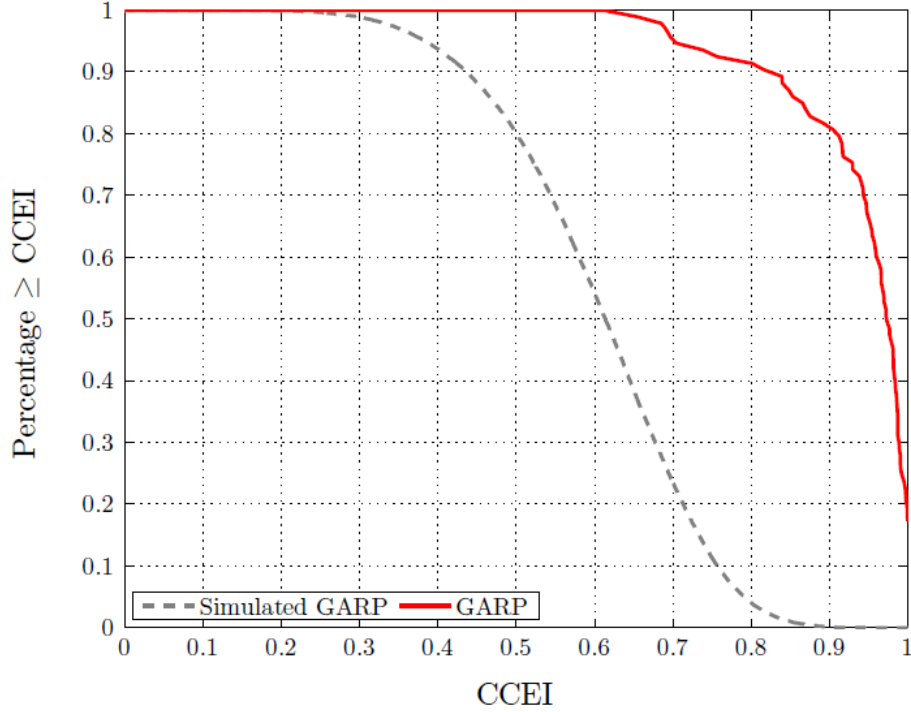


Figure 3.2: CCEI Source: Quah

### 3.24 Measurement Error

A similar experiment of budget allocation but using 3 goods instead of 2, is in Ahn, D., Choi, S., Gale, D., & Kariv, S. (2014) (Estimating ambiguity aversion in a portfolio choice experiment. *Quantitative Economics*, 5(2), 195–223). This experiment has  $n = 154$  individuals, and  $T = 50$  trials.

- The GARP test is successful 12.99% of cases (individuals).
- Aguiar and Kashaev (2020, Restud) extend GARP to allow for measurement error.
- Measurement error can be either present in consumption  $x_t = x_t^* + w_t^c$  or in prices  $p_t = p_t^* + w_t^p$  for  $x_t^*, p_t^*$  being the correctly measured prices and consumption.
- GARP should hold for the (ideal), well-measured dataset  $(p_t^*, x_t^*)_{t=1}^T$  but not necessarily for the mismeasured dataset  $(p_t, x_t)_{t=1}^T$ .
- Without any knowledge about measurement error then rationality becomes untestable.

- We assume that measurement error is random across individuals and that the expectation is equal to zero:  $E[\mathbf{w}_t^e] = 0$ ,  $E[\mathbf{w}_t^p] = 0$ .
- These assumptions are backed up by validation experimental data that shows that consumers may incur in trembling hand errors, or perception errors that are not systematic (see our paper).
- The key is that we do not know the distribution of measurement error, we only know it's average behavior.
- Aguiar and Kashaev (2020) provide a methodology to test GARP in the ideal (correctly measured data) with knowledge about average behavior of measurement error.
- They conclude that the null hypothesis that every individual in this experiment is consistent with GARP cannot be rejected at the 95 confidence level. They also find that misperception of prices seems to be the culprit of spurious violations of rationality.
- Does this confirm rationality? NO!
- A theory cannot be confirmed by this exercise alone, but current evidence is insufficient to reject rationality (at least in the context of this simple experiment).

### 3.25 Welfare Evaluation

We will compare the wealth needed to achieve a certain level of utility at a given price-wealth pair.

$$u^1 = v(p^1, w)$$

$$u^0 = v(p^0, w).$$

We are interested in comparing the expenditure needed to achieve  $u^1$  or  $u^0$ .

For a fixed reference price and a given utility level  $u$ , the needed expenditure to reach it is given by

$$e(p^{ref}, u).$$

Thus the comparison in terms of expenditure is given by

$$e(p^{ref}, u^1) - e(p^{ref}, u^0)$$

$$e(p^{ref}, v(p^1, w)) - e(p^{ref}, v(p^0, w)).$$

This expression will be positive when it takes more wealth to achieve  $u^1$  at prices  $p^{ref}$  than to achieve  $u^0$ . This difference is measured in dollar terms. That is why  $e(p^{ref}, v(p, w))$  is often called money metric indirect utility function.

We can construct many money metric indirect utility functions with different reference price.

When  $p^{ref} = p^0$  is the original price situation, the expenditure change equals the wealth change that makes the consumer indifferent between the new price with the old wealth and the old price with the new wealth. This asks what wealth change is equivalent to the price change.

The **equivalent variation** is

$$EV(p^0, p^1, w) = e(p^0, v(p^1, w)) - e(p^0, v(p^0, w)) = e(p^0, v(p^1, w)) - w,$$

since  $e(p^0, v(p^0, w)) = w$ .

With this sign convention, equivalent variation is the money change at the reference price that is equivalent to the price change: it is willingness to pay for an improvement, and its negative is willingness to pay to avoid a worsening.

The other case considered is the one where the new price is taken as the reference price, when  $p^{ref} = p^1$ , the expenditure change equals the wealth change that makes the consumer indifferent between the original situation  $(p^0, w)$  and the new situation  $(p^1, w + \Delta w)$ .

The **compensating variation** is defined as:

$$CV(p^0, p^1, w) = e(p^1, v(p^1, w)) - e(p^1, v(p^0, w)) = w - e(p^1, v(p^0, w)).$$

The sign convention for compensating variation is analogous: a negative value for a worsening is the compensation required to restore the initial utility.

### 3.26 Homothetic and Quasilinear Utility Functions

**Definition 18.** Homothetic preferences. A preference relation is homothetic if  $x \succeq y$  implies  $\alpha x \succeq \alpha y$  for every  $\alpha > 0$  whenever the scaled bundles are feasible. Under the usual continuity

and monotonicity assumptions, it admits a positively homogeneous utility representation; an arbitrary representation need not itself be homogeneous.

The Cobb--Douglas representation  $u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$  is a standard example.

**Definition 19.** Quasilinear preferences. A numeraire good  $L$  is quasilinear when preferences admit a representation

$$u(x) = g(x_1, \dots, x_{L-1}) + x_L.$$

Equivalently, translating both bundles by the same amount of the numeraire does not change their ranking whenever both translated bundles remain feasible.

## Part II

# Choice Beyond the Benchmark Model

## Chapter 4

# Behavioral Economics: Sparse-Max and Rational Inattention

The benchmark consumer maximizes utility using the true budget and prices. A sparse-max consumer keeps the true budget constraint but forms a decision using a perceived price vector. This chapter states the behavioral rule explicitly and records which benchmark conclusions no longer follow automatically.

### 4.1 The Sparse-Max Decision Rule

Let  $X = \mathbb{R}_+^L$ , true prices  $p \gg 0$ , perceived prices  $\hat{p} \gg 0$ , and wealth  $w > 0$ . Assume  $u$  is differentiable and that the tangency system below has a solution. A sparse-max choice is obtained by first selecting a bundle that satisfies the perceived-price first-order condition

$$\nabla u(x) = \lambda \hat{p}, \quad \lambda > 0,$$

and then choosing  $\lambda$  so that the selected bundle satisfies the actual budget equation  $p \cdot x = w$ . At a boundary solution, replace the equality by the corresponding KKT inequalities. This two-stage rule is written  $x^s(p, \hat{p}, w)$ . It is not the ordinary utility-maximization problem with  $\hat{p}$  in the budget constraint.

When the rational demand  $x^r(\hat{p}, \tilde{w})$  is single-valued and locally invertible in wealth, the

same sparse-max bundle can be written as  $x^s(p, \hat{p}, w) = x^r(\hat{p}, \tilde{w})$  for the unique  $\tilde{w}$  satisfying  $p \cdot x^r(\hat{p}, \tilde{w}) = w$ . This is a change of representation, not an assertion that the consumer faces  $\hat{p}$  in the market.

## 4.2 Cobb--Douglas Sparse-Max Demand

**Example 6.** Let  $u(x) = \sum_{\ell=1}^L \alpha_{\ell} \log x_{\ell}$  with  $\alpha_{\ell} > 0$  and  $\sum_{\ell} \alpha_{\ell} = 1$ . The perceived-price first-order conditions give

$$x_{\ell} = \frac{\alpha_{\ell}}{\lambda \hat{p}_{\ell}}.$$

Imposing the actual budget yields  $\lambda = \frac{1}{w} \sum_{k=1}^L \alpha_k \frac{p_k}{\hat{p}_k}$  and therefore

**Example 7.**

$$x_{\ell}^s(p, \hat{p}, w) = \frac{\alpha_{\ell} w / \hat{p}_{\ell}}{\sum_{k=1}^L \alpha_k p_k / \hat{p}_k}.$$

Indeed,  $\sum_{\ell} p_{\ell} x_{\ell}^s = w$ . If  $\hat{p} = p$ , this reduces to the ordinary Cobb--Douglas demand  $x_{\ell} = \alpha_{\ell} w / p_{\ell}$ .

## 4.3 Perceived Prices and Limited Attention

One parsimonious attention rule combines the true price with a default price  $p^d \gg 0$ :

$$\hat{p}_{\ell}(p) = m_{\ell} p_{\ell} + (1 - m_{\ell}) p_{\ell}^d, \quad m_{\ell} \in [0, 1].$$

The vector  $m = (m_1, \dots, m_L)$  measures attention to price components. Full attention is  $m_{\ell} = 1$  for every  $\ell$ ; setting  $m_{\ell} = 0$  makes that component equal to its default. With  $M = \text{diag}(m_1, \dots, m_L)$ ,  $\hat{p}(p) = Mp + (I - M)p^d$ .

**Proposition 17.** *Comparative statics under perceived prices. Suppose the tangency bundle can be written as a differentiable map  $r(\hat{p}, \lambda)$  and  $x^s(p, w) = r(\hat{p}(p), \lambda(p, w))$ , where  $\lambda(p, w)$  is determined by  $p \cdot r(\hat{p}(p), \lambda) = w$ . Then the chain rule gives*

$$D_p x^s = D_{\hat{p}} r M + \partial_{\lambda} r (\nabla_p \lambda)^{\top}, \quad D_w x^s = \partial_{\lambda} r \partial_w \lambda.$$

The second terms are the response of the multiplier needed to restore the actual budget. Consequently, an identity such as  $S^s = S^r M$  requires additional compensation and normalization

assumptions; it is not a generic implication of the attention rule.

Because the default price is fixed, scaling  $(p, w)$  by a common positive factor generally changes  $\hat{p}(p)$  and therefore can violate homogeneity of degree zero. WARP, Slutsky symmetry, and welfare theorems must likewise be checked for the specified behavioral demand rather than imported from the rational benchmark.

## 4.4 Endogenizing Attention

Let the unknown state be  $X \sim \mathcal{N}(x^d, \sigma_x^2)$  and let the signal be  $S = X + \varepsilon$  with independent  $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ . Under quadratic loss  $\ell(a, X) = \frac{1}{2}(a - X)^2$ , the optimal action is the posterior mean

$$a^*(s) = \mathbb{E}[X \mid S = s] = ms + (1 - m)x^d, \quad m = \frac{\sigma_x^2}{\sigma_x^2 + \sigma_\varepsilon^2} \in [0, 1].$$

The posterior variance is  $(1 - m)\sigma_x^2$ , so expected loss at the optimal action is  $\frac{1}{2}(1 - m)\sigma_x^2$ . If attention itself is costly, represent the cost by an increasing function  $C(m)$  and solve

$$\max_{0 \leq m < 1} -\frac{1}{2}(1 - m)\sigma_x^2 - C(m).$$

For example, with  $C(m) = \kappa m^2/2$ , an interior optimum satisfies  $m = \sigma_x^2/(2\kappa)$  and is truncated to  $[0, 1]$  when the unconstrained value lies outside that interval. The comparative statics depend on the cost specification.

A capacity formulation is also transparent. If the information cost is  $-\frac{1}{2}\log(1 - m)$  and capacity is bounded by  $K$ , then  $-\frac{1}{2}\log(1 - m) \leq K$  implies  $m \leq 1 - e^{-2K}$ . Since the value of information is increasing in  $m$ , the capacity-constrained optimum attains this upper bound.

## 4.5 Discrimination with Rational Inattention

Consider a firm deciding whether to hire an applicant. Productivity is  $Q$  and hiring yields payoff  $Q$ , while rejecting yields 0. Before observing an applicant-specific signal, the firm observes  $G \in \{R, B\}$  and has prior  $Q \mid G \sim \mathcal{N}(q_G, \sigma_q^2)$  with  $q_R < q_B$ .

The firm observes  $Y = Q + \varepsilon$ , where  $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$  is independent. The posterior mean is

$$m_G(y) = \mathbb{E}[Q \mid Y = y, G] = \alpha y + (1 - \alpha)q_G, \quad \alpha = \frac{\sigma_q^2}{\sigma_q^2 + \sigma_\varepsilon^2}.$$

**Proposition 18.** *Optimal hiring rule. Conditional on  $(y, G)$ , hire if and only if  $m_G(y) > 0$ ; otherwise reject. For  $\alpha > 0$  the signal threshold is  $y_G = -(1 - \alpha)q_G/\alpha$ .*

If both group means are negative, the lower-mean group has the higher threshold and is hired less often under a common signal distribution. If both means are positive, the same threshold formula applies, but group-specific rejection rates depend on the signal distribution. Choosing the noise level solves  $\max_{\sigma_\varepsilon^2 \geq 0} \mathbb{E}[\max\{m_G(Y), 0\}] - C(\sigma_\varepsilon^2)$ ; group-specific attention requires the group priors and a fully specified cost function.

## Chapter 5

# Choice under Uncertainty

### 5.1 Primitives

Let  $Z \subseteq \mathbb{R}$  be a finite set of monetary prizes. A lottery is a probability mass function  $p : Z \rightarrow [0, 1]$  satisfying  $\sum_{z \in Z} p(z) = 1$ . Write  $\Delta(Z)$  for the set of lotteries and  $\delta_z$  for the degenerate lottery assigning probability one to  $z$ . Mixtures are defined pointwise by

$$[\alpha p + (1 - \alpha)q](z) = \alpha p(z) + (1 - \alpha)q(z), \quad \alpha \in [0, 1].$$

For a Bernoulli utility  $u : Z \rightarrow \mathbb{R}$ , expected utility is  $U(p) = \sum_{z \in Z} u(z)p(z)$ .

### 5.2 Stochastic dominance

**Definition 20.** First-order stochastic dominance. For lotteries  $p, q \in \Delta(Z)$  with cumulative distribution functions  $F_p(t) = \sum_{z \leq t} p(z)$  and  $F_q(t) = \sum_{z \leq t} q(z)$ , write  $p \succeq^{FOSD} q$  if  $F_p(t) \leq F_q(t)$  for every  $t \in \mathbb{R}$ . The relation is generally incomplete.

**Theorem 4.** If  $p \succeq^{FOSD} q$  and  $u$  is increasing, then  $U(p) \geq U(q)$  for expected utility  $U$ . Strict inequality follows when the dominance is strict on a set that receives positive probability and  $u$  is strictly increasing there.

### 5.3 Model and expected utility

We observe a choice correspondence on a family of subsets of  $\Delta(Z)$  and represent the decision maker's preference by a relation  $\succeq$  on lotteries.

**Definition 21.** Expected utility representation. The preference relation has an expected utility representation if there is a Bernoulli utility  $u : Z \rightarrow \mathbb{R}$  such that  $p \succeq q \iff \sum_{z \in Z} u(z)p(z) \geq \sum_{z \in Z} u(z)q(z)$ .

## 5.4 Independence Axiom

We are assuming  $\succeq$  are rational.

**Axiom 4.** (*Independence Axiom*) For all  $p, q, r \in \Delta(Z)$  and  $\alpha \in [0, 1]$ ,  $p \succeq q$  implies  $\alpha p + (1 - \alpha)r \succeq \alpha q + (1 - \alpha)r$ .

The other axiom that we need is more technical

**Axiom 5.** (*Mixture continuity*) For all  $p, q, r \in \Delta(Z)$ , if  $p \succ q \succ r$ , then the set  $\{\alpha \in [0, 1] | \alpha p + (1 - \alpha)r \succ q\}$  and  $\{\alpha \in [0, 1] | q \succ \alpha p + (1 - \alpha)r\}$  are open in  $[0, 1]$ .

## 5.5 Representation Theorem

**Theorem 5.** The rational preference relation  $\succeq$  satisfies the Independence Axiom and Mixture continuity if and only if there exists a (Bernoulli) utility function  $u : Z \rightarrow \mathbb{R}$  such that:  $p \succeq q$  if and only if  $\sum_{z \in Z} u(z)p(z) \geq \sum_{z \in Z} u(z)q(z)$ .

First we prove that there is an affine representation of  $\succeq$ .

**Lemma 2.** If the preference relation  $\succeq$  satisfies the Independence Axiom and mixture continuity then there is an affine function  $U : \Delta(Z) \rightarrow \mathbb{R}$  that represents  $\succeq$ .

*Proof.* Choose best and worst degenerate lotteries  $p^+ = \delta_{z^+}$  and  $p^- = \delta_{z^-}$  among the finitely many prizes. Repeated use of independence shows  $p^+ \succeq p \succeq p^-$  for every lottery  $p$ . If  $p^+ \sim p^-$ , all lotteries are indifferent and a constant affine representation works. Otherwise, independence implies

$$\beta p^+ + (1 - \beta)p^- \succ \alpha p^+ + (1 - \alpha)p^- \iff \beta > \alpha.$$

For a fixed  $p$ , mixture continuity makes the two sets of mixture weights giving an outcome weakly above or weakly below  $p$  closed. They cover the connected interval  $[0, 1]$  and contain opposite endpoints, so they intersect. Strict monotonicity along the best--worst segment gives a unique  $\alpha_p$  with

$$p \sim \alpha_p p^+ + (1 - \alpha_p)p^-.$$

Thus  $U(p) = \alpha_p$  represents  $\succeq$ . Finally, independence permits substitution of the equivalent best--worst mixtures for  $p$  and  $q$ , yielding

$$U(\beta p + (1 - \beta)q) = \beta U(p) + (1 - \beta)U(q).$$

Hence  $U$  is affine on the lottery simplex.

□

The second step is to show that an affine representation on the lottery simplex has the expected-utility form.

**Lemma 3.** *Let  $U : \Delta(Z) \rightarrow \mathbb{R}$ . Then  $U$  is affine if and only if there exists a  $u : Z \rightarrow \mathbb{R}$  such that  $U(p) = \sum_{z \in Z} u(z)p(z)$ .*

For finite  $Z$ , no functional-analytic representation theorem is needed. Define  $u(z) = U(\delta_z)$ . Since every lottery has the finite decomposition  $p = \sum_{z \in Z} p(z)\delta_z$ , affinity of  $U$  gives  $U(p) = \sum_{z \in Z} u(z)p(z)$ . The converse follows immediately from distributivity.

**Example 8.**  $Z = \{0, 10\}$   $p \in \Delta(Z)$   $p(0) = \frac{1}{2}$ ,  $p(10) = \frac{1}{2}$ .  $U(p) = \frac{1}{2}u(10) + \frac{1}{2}u(0)$ , I can choose the Bernoulli utility to be:

$$u(10) = \kappa \cdot \log(10), \kappa > 0.$$

## 5.6 Recoverability

**Theorem 6.** *Let  $U : \Delta(Z) \rightarrow \mathbb{R}$  and  $U' : \Delta(Z) \rightarrow \mathbb{R}$  be affine maps that represent  $\succeq$  on  $\Delta(Z)$ , then there exist real numbers  $\alpha > 0$  and  $b$  such that  $U'(p) = \alpha U(p) + b$  for all  $p \in \Delta(Z)$ . If  $u, v$  are Bernoulli utilities such that  $u \cdot p$  and  $v \cdot p$  represent  $\succeq$  then there are real numbers  $\alpha > 0$  and  $b$  such that  $v \cdot p = \alpha u \cdot p + b$ .*

Think about this. How it differs from the recoverability result of the common rational preference  $\succeq$ .

## 5.7 Attitudes towards Risk

Assume that the certainty set contains the convex hull of the finite prize set. For a lottery  $p$ , its expected monetary value is  $\bar{z}(p) = \sum_{z \in Z} zp(z)$  and the corresponding certain outcome is  $\delta_{\bar{z}(p)}$ .

**Definition 22.** Risk aversion. A decision maker is risk averse if  $\delta_{\bar{z}(p)} \succeq p$  for every lottery  $p$ . Under expected utility, risk aversion is equivalent to concavity of the Bernoulli utility:

$$\sum_{z \in Z} p(z)u(z) \leq u\left(\sum_{z \in Z} p(z)z\right).$$

Certainty equivalent and risk premium. If  $u$  is strictly increasing and its range contains  $U(p)$ , the certainty equivalent is the prize  $CE(p, u) = u^{-1}(U(p))$  and the (monetary) risk premium is  $\rho(p, u) = \bar{z}(p) - CE(p, u)$ . A risk-averse expected-utility decision maker has  $\rho(p, u) \geq 0$ .

Probability premium. For prizes  $x - \varepsilon, x, x + \varepsilon$  in the certainty set,  $\pi(x, \varepsilon, u)$  is defined by

$$u(x) = \left(\frac{1}{2} + \pi\right)u(x + \varepsilon) + \left(\frac{1}{2} - \pi\right)u(x - \varepsilon).$$

It is the change in probability needed to make the symmetric two-point lottery equivalent to the sure prize.

**Proposition 19.** For a twice differentiable, strictly increasing Bernoulli utility, the local risk premium for a symmetric small spread satisfies

$$\rho\left(\frac{1}{2}\delta_{x-\varepsilon} + \frac{1}{2}\delta_{x+\varepsilon}, u\right) = -\frac{u''(x)}{2u'(x)}\varepsilon^2 + o(\varepsilon^2).$$

Thus local concavity corresponds to a nonnegative local risk premium.

## 5.8 Expected Utility for Infinite Sets

Let  $Z$  be a set of prizes, we say that the collection of subsets  $\Sigma \subseteq 2^Z$  is a  $\sigma$ -algebra when

- $\emptyset \in \Sigma$
- if  $E \in \Sigma$  then  $E^c \in \Sigma$
- If  $\{A_n\}$  is a countable collection of sets such that  $A_n \in \Sigma$  then  $\cup_n A_n \in \Sigma$ .

We say that  $(Z, \Sigma)$  is a measurable space whenever  $\Sigma$  is a  $\sigma$ -algebra.

**Definition 23.** We say that  $p : \Sigma \rightarrow [0, 1]$  is a probability measure if:

- $p(Z) = 1$
- if  $\{A_n\}$  is a countable collection of pairwise disjoint sets such that  $A_n \in \Sigma$ , then  $p(\cup_n A_n) = \sum_n p(A_n)$ .

We call the collection of probability measures  $\Delta(Z, \Sigma)$  for short  $\Delta(Z)$  when  $\Sigma$  is fixed.

$\alpha p + (1 - \alpha)q \in \Delta(Z)$  is defined as  $(\alpha p + (1 - \alpha)q)(E) = \alpha p(E) + (1 - \alpha)q(E)$ .

We fix a topology on this space:  $p^n \xrightarrow{w^*} p$  under the weak\* topology if  $\int f dp^n \rightarrow \int f dp$  for continuous and bounded functions  $f$ .

**Axiom 6.** (*Weak-star continuity*) For any  $p \in \Delta(Z)$  the sets  $\{q \in \Delta(Z) : q \succ p\}$  and  $\{q \in \Delta(Z) : p \succ q\}$  are open in the weak\* topology.

**Theorem 7.** Let  $Z$  be a compact metric prize space and  $\Delta(Z)$  the Borel probability measures with the weak-star topology. A rational preference relation  $\succeq$  on  $\Delta(Z)$  satisfies the independence axiom and weak-star continuity if and only if there exists a  $u : Z \rightarrow \mathbb{R}$  bounded and continuous such that  $U(p) = \int u(z)dp(z)$  represents  $\succeq$ .

## 5.9 Application: Demand for a Risky Asset

An asset is a divisible claim to a financial return in the future.

Let  $\alpha, \beta$  denote the amounts of money invested in a risky and safe asset respectively. The risky asset pays a random return of  $Z$  dollars per dollar invested and the safe asset pays 1 dollar with certainty per dollar invested. The random return has a distribution  $F(z)$  (CDF). We assume that  $\int z dF(z) > 1$ , or that the risky asset expected return exceeds the return of the safe asset. Then the total return is  $\alpha Z + \beta$ . Of course, we must have that  $\alpha + \beta = w$ .

$$\max_{\alpha, \beta \geq 0} \int u(\alpha z + \beta) dF(z)$$

$$s.t. \quad \alpha + \beta = w.$$

Equivalently we want to solve.

$$\max_{0 \leq \alpha \leq w} \int u(w + \alpha(z - 1)) dF(z).$$

The first-order conditions are:

$$\phi(\alpha^*) = \int u'(w + \alpha^*[z - 1])[z - 1] dF(z) \begin{cases} \leq 0 & \text{if } \alpha^* < w \\ \geq 0 & \text{if } \alpha^* > 0. \end{cases}$$

Note that  $\int z dF(z) > 1$  implies that  $\phi(0) > 0$ .

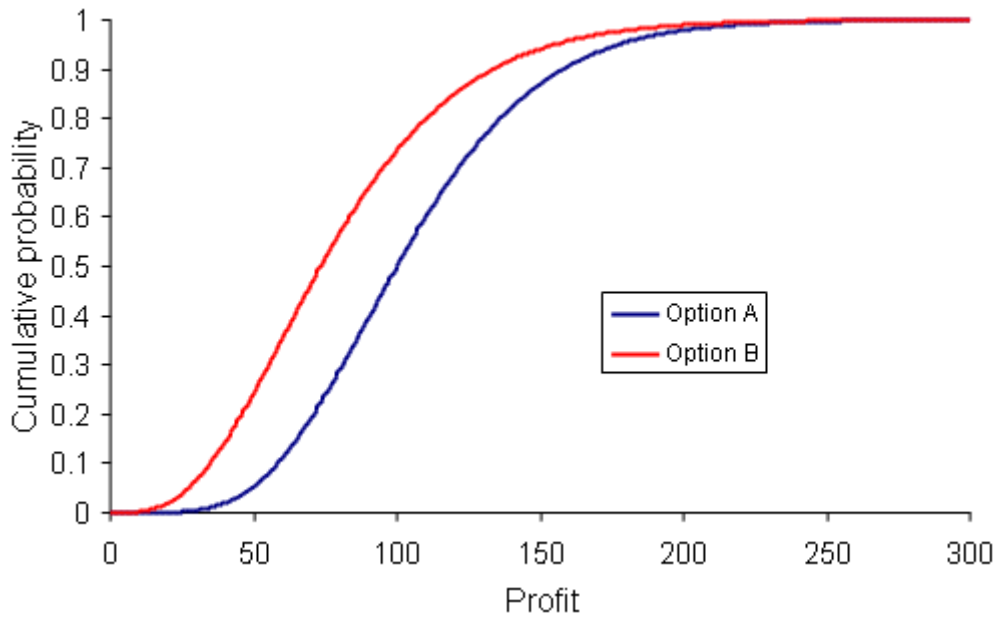


Figure 5.1: First order stochastic dominance. Option A dominates Option B.

Because  $u'(w) > 0$ .

And  $\phi(0) = \int u'(w)[z - 1]dF(z)$ .

So  $\alpha^* = 0$  cannot satisfy the first order conditions.

If a risk is actuarially favorable, then a risk-averse expected-utility maximizer will accept at least a sufficiently small amount under the stated differentiability and interior-wealth assumptions.

## 5.10 Dominance

**Theorem 8.** *First-order stochastic dominance. If  $F \succeq^{FOSD} G$ , then  $\int u dF \geq \int u dG$  for every increasing Bernoulli utility  $u$  for which the integrals exist. Strict dominance together with strict monotonicity yields a strict inequality under the usual support conditions.*

$F \succeq^{FOSD} G$  FOSD means  $F(x) \leq G(x)$  for all  $x \in Z$  with strict inequality for some  $x \in Z$ .

## 5.11 The Allais Paradox

$Z$  is finite in fact  $N = 3$ ,

| First Prize  | Second Prize | Third Prize |
|--------------|--------------|-------------|
| 2.5 millions | 0.5 millions | 0           |

Two choice sets:

The first,

$L_1 = (0, 1, 0)$  and  $L'_1 = (0.10, 0.89, 0.01)$ .

The second,

$L_2 = (0, 0.11, 0.89)$  and  $L'_2 = (0.10, 0, 0.90)$ .

Some individuals choose  $L_1 \succ L'_1$  and  $L'_2 \succ L_2$ .

This is inconsistent with expected utility.

If  $L_1 \succ L'_1$  then

$$u_{0.5} > (0.10)u_{2.5} + (0.89)u_{.5} + (0.01)u_0.$$

Adding,  $(0.89)u_0 - (0.89)u_{0.5}$  to both sides we get,

$$(0.11)u_{0.5} + (0.89)u_0 > (0.1)u_{2.5} + (0.9)u_0,$$

and thus  $L_2 \succ L'_2$ .

## 5.12 Revealed Preference of Expected Utility under Concavity

Let  $S$  be a finite set of states with fixed probabilities  $\pi_s > 0$ , and let a contingent consumption plan be  $x = (x_s)_{s \in S} \in \mathbb{R}_{++}^S$ . At prices  $p \in \mathbb{R}_{++}^S$  and wealth  $w > 0$ , the consumer solves  $\max_x \sum_{s \in S} \pi_s u(x_s)$  subject to  $\sum_{s \in S} p_s x_s \leq w$ , where  $u$  is continuous, increasing and concave.

**Proposition 20.** *Interior first-order conditions. If  $u$  is differentiable and an optimum is interior, then there is  $\lambda > 0$  such that  $\pi_s u'(x_s) = \lambda p_s$  for every  $s \in S$ . Concavity makes these conditions sufficient for a global optimum; boundary observations require the corresponding KKT inequalities.*

**Definition 24.** Risk-averse expected-utility rationalization. A finite dataset  $O^T = \{(p^t, x^t)\}_{t=1}^T$  is rationalized if there is one continuous, increasing and concave Bernoulli utility  $u$  such that each  $x^t$  maximizes  $\sum_s \pi_s u(x_s)$  on the budget set  $\{x : p^t \cdot x \leq p^t \cdot x^t\}$ .

**Theorem 9.** *Finite-data test (cited). Assume all observations are interior and  $\pi_s > 0$ . The dataset has a risk-averse expected-utility rationalization if and only if there are numbers  $\beta_s^t > 0$  and  $\lambda^t > 0$  such that  $\pi_s \beta_s^t = \lambda^t p_s^t$  for every  $s, t$  and  $x_s^t > x_{s'}^{t'} \implies \beta_s^t \leq \beta_{s'}^{t'}$  for every pair of observed state-contingent quantities. The monotonicity condition is the finite signature of a single decreasing marginal-utility function; the price equations are the common first-order conditions.*

The theorem is an Afriat-style interpolation result for one Bernoulli utility across states and observations. It is stated for interior data to keep the notation transparent; zero quantities are handled by replacing equalities with KKT inequalities.

### 5.13 Experimental Evidence

Polisson and Quah develop a revealed preferences test for expected utility. They apply their framework to the famous Choi, Fisman, Gale and Kariv (2007) dataset and find the following passing rates.

- GARP is the benchmark revealed-preference test.
- RDEU denotes rank-dependent expected utility.
- EU denotes expected utility.
- EU\* imposes concavity of the Bernoulli utility function in addition to expected utility.

The “drop” column reports how many observations the test permits the researcher to discard.

|        | GARP        | RDEU        | EU          | EU*       |
|--------|-------------|-------------|-------------|-----------|
| Drop 0 | 12/47 (26%) | 2/47 (4%)   | 2/47 (4%)   | 0/47 (0%) |
| Drop 1 | 14/47 (30%) | 7/47 (15%)  | 5/47 (11%)  | 0/47 (0%) |
| Drop 2 | 27/47 (57%) | 9/47 (19%)  | 7/47 (15%)  | 2/47 (4%) |
| Drop 3 | 32/47 (68%) | 12/47 (26%) | 10/47 (21%) | 2/47 (4%) |

Table 1: Results

Figure 5.2: Share of subjects passing each revealed-preference test

### 5.14 Measuring Risk Aversion in the Lab

The Holt and Laury price list method.

| Table 1: Holt and Laury (2002) |                               |                      |                               |                      |                                                 |                               |                                |
|--------------------------------|-------------------------------|----------------------|-------------------------------|----------------------|-------------------------------------------------|-------------------------------|--------------------------------|
| Row No.                        | Option A Outcome A 1 = \$2.00 | Outcome A 2 = \$1.60 | Option B Outcome B 1 = \$3.85 | Outcome B 2 = \$0.10 | RRA if row was last choice of A and below all B | Proportion choices payoffs x1 | Proportion choices payoffs x20 |
| 1                              | Prob. 1/10                    | Prob. 9/10           | Prob. 1/10                    | Prob. 9/10           | [-1, 71; -0.95]                                 | 0.01                          | 0.01                           |
| 2                              | Prob. 2/10                    | Prob. 8/10           | Prob. 2/10                    | Prob. 8/10           | [-0.95; -0.49]                                  | 0.01                          | 0.01                           |
| 3                              | Prob. 3/10                    | Prob. 7/10           | Prob. 3/10                    | Prob. 7/10           | [-0.49; -0.14]                                  | 0.06                          | 0.04                           |
| 4                              | Prob. 4/10                    | Prob. 6/10           | Prob. 4/10                    | Prob. 6/10           | [-0.14; 0.15]                                   | 0.26                          | 0.13                           |
| 5                              | Prob. 5/10                    | Prob. 5/10           | Prob. 5/10                    | Prob. 5/10           | [0.15; 0.41]                                    | 0.26                          | 0.19                           |
| 6                              | Prob. 6/10                    | Prob. 4/10           | Prob. 6/10                    | Prob. 4/10           | [0.41; 0.68]                                    | 0.23                          | 0.23                           |
| 7                              | Prob. 7/10                    | Prob. 3/10           | Prob. 7/10                    | Prob. 3/10           | [0.68; 0.97]                                    | 0.13                          | 0.22                           |
| 8                              | Prob. 8/10                    | Prob. 2/10           | Prob. 8/10                    | Prob. 2/10           | [0.97; 1.37]                                    | 0.03                          | 0.11                           |
| 9                              | Prob. 9/10                    | Prob. 1/10           | Prob. 9/10                    | Prob. 1/10           | [1.37; ∞)                                       | 0.01                          | 0.06                           |
| 10                             | Prob. 10/10                   | Prob. 0/10           | Prob. 10/10                   | Prob. 0/10           | non-monotone                                    | 0.00                          | 0.00                           |

Figure 5.3: Price list method for eliciting risk aversion

Individuals choose between  $A$  and  $B$  in each of the ten rows.

There are two outcomes for each option  $A_1, A_2$  and  $B_1, B_2$  with changing probabilities for each row.

The expected value of  $A$  is higher for the first four rows, and it switches to lower for the last six rows.

- A risk-neutral subject chooses  $A$  from  $t = 1, \dots, 4$  switch at 5 and remain there.

- Option  $B$  has higher variance:

- An individual who switches to  $B$  between row 6 and row 10 is classified as being risk-averse and a more risk-averse individual typically switches later, under the maintained utility and payoff assumptions.

At random you pay one of the rows. To model a decision-maker's incentives,

## Part III

# Aggregation and Random Choice

## Chapter 6

# Aggregate Demand

### 6.1 Three questions about aggregation

- When does aggregate demand depend on prices and total wealth rather than on the entire wealth distribution?
- When does aggregate demand satisfy a demand restriction such as WARP?
- When can aggregate demand be given a welfare interpretation?

Let consumers be indexed by  $i = 1, \dots, I$ . Consumer  $i$  has Marshallian demand  $x_i(p, w_i)$  at common prices  $p \in \mathbb{R}_{++}^L$  and wealth  $w_i > 0$ . Aggregate demand is  $X(p, \mathbf{w}) = \sum_{i=1}^I x_i(p, w_i)$ , where  $\mathbf{w} = (w_1, \dots, w_I)$ . Total wealth is  $W = \sum_i w_i$ . Without additional structure, two wealth distributions with the same  $W$  can generate different aggregate bundles.

### 6.2 Gorman aggregation

**Definition 25.** Gorman form. Consumer  $i$  has Gorman-form indirect utility if  $v_i(p, w_i) = a_i(p) + b(p)w_i$  for a common function  $b(p) > 0$  and consumer-specific functions  $a_i(p)$ .

**Proposition 21.** *Gorman aggregation (cited). Suppose each  $v_i$  has the Gorman form and is differentiable. Roy's identity gives  $x_i(p, w_i) = -\frac{\nabla_p a_i(p) + w_i \nabla_p b(p)}{b(p)}$ . Therefore  $X(p, \mathbf{w}) = A(p) + B(p)W$ , where  $A(p) = -\sum_i \nabla_p a_i(p)/b(p)$  and  $B(p) = -\nabla_p b(p)/b(p)$ . Aggregate demand consequently depends on  $(p, W)$  and not on the distribution of wealth.*

Under the usual differentiability and integrability conditions, a common wealth derivative  $D_{w_i} x_i(p, w_i) = B(p)$  is the local signature of Gorman aggregation. The converse is a local

statement and should not be read as a global representation without the regularity assumptions.

### 6.3 Aggregate demand and WARP

Even if every individual demand satisfies WARP, aggregate demand need not do so. Aggregate WARP compares two price--total-wealth pairs while holding fixed the rule that assigns total wealth across consumers; the corresponding compensation need not leave each consumer's wealth unchanged.

**Definition 26.** Aggregate WARP. A single-valued aggregate demand  $X(p, W)$  satisfies WARP if, for distinct observations  $(p, W)$  and  $(p', W')$ ,  $p \cdot X(p', W') \leq W$  implies  $p' \cdot X(p, W) > W'$  whenever  $X(p, W) \neq X(p', W')$ .

**Proposition 22.** *Sufficient condition for aggregate WARP. Suppose wealth shares are fixed,  $w_i = \alpha_i W$  with  $\alpha_i \geq 0$  and  $\sum_i \alpha_i = 1$ , and suppose each map  $p \mapsto x_i(p, \alpha_i W)$  satisfies the weak uncompensated law of demand on a convex price domain. Then  $X(p, W) = \sum_i x_i(p, \alpha_i W)$  satisfies the weak uncompensated law of demand. With Walras' law and the usual strictness condition, aggregate WARP follows.*

The proof is additive: sum the individual inequalities at the common wealth shares. If the shares change with prices or if wealth is reallocated after compensation, this argument no longer applies.

### 6.4 Representative consumers and welfare

**Definition 27.** Positive representative consumer. A preference relation  $\succeq^{rep}$  is a positive representative consumer for  $X$  if its Walrasian demand equals  $X(p, W)$  for every price--wealth pair in the domain. This is a statement about demand representation, not about welfare.

Normative representative consumer. Fix a social welfare function  $\mathcal{W} : \mathbb{R}^I \rightarrow \mathbb{R}$  and individual indirect utilities  $v_i$ . A wealth allocation  $\mathbf{w}(p, W)$  is welfare-optimal if it solves  $\max_{\mathbf{w}} \mathcal{W}(v_1(p, w_1), \dots, v_I(p, w_I))$  subject  $W$ . A positive representative consumer is normative relative to  $\mathcal{W}$  only when the wealth rule used to form  $X$  is welfare-optimal at every  $(p, W)$ .

Gorman aggregation can deliver a positive representative consumer, but it does not select a social welfare function or justify interpersonal utility comparisons. Normative claims require the welfare criterion and the wealth-allocation rule to be stated explicitly.

## Chapter 7

# Discrete Choice and Random Utility

### 7.1 Choice from a finite menu

There are alternatives  $j \in \{0, 1, \dots, J\}$ , where  $j = 0$  is the outside option. Consumer  $i$  chooses one alternative from the menu after comparing conditional indirect utilities  $U_{ij}^* = V_j(x_j, p_j, y_i) + \varepsilon_{ij}$ . Here  $x_j$  denotes observed attributes,  $V_j$  is mean utility, and  $\varepsilon_{ij}$  is an unobserved preference shock. The choice probability is  $P_{ij} = \Pr\{U_{ij}^* \geq U_{ik}^* \text{ for every } k\}$ , with strict inequalities almost surely when the shock distribution is continuous.

If the outside good has price  $p_0$  and quantity  $z$ , a budget identity  $p_j + p_0 z = y_i$  yields  $z = (y_i - p_j)/p_0$  for a purchased alternative. This reduction is valid only when the implied outside-good quantity is feasible.

### 7.2 Multinomial logit and IIA

Suppose the shocks are independent and identically distributed Type-I extreme value with scale  $\mu > 0$ . Then the multinomial-logit probabilities are  $P_{ij} = \frac{\exp(V_j/\mu)}{\sum_{k=0}^J \exp(V_k/\mu)}$ .

**Proposition 23.** *Independence of irrelevant alternatives. Under multinomial logit, for any two alternatives  $j, k$  that remain available,  $\frac{P_{ij}}{P_{ik}} = \exp((V_j - V_k)/\mu)$  is independent of the other alternatives in the menu.*

**Example 9.** Red-bus/blue-bus. If walking and a red bus initially have equal probabilities, adding a blue bus that is nearly identical to the red bus should split the bus share while leaving walking's

share close to its original value. Multinomial logit instead preserves the walking-to-red-bus odds ratio, illustrating the empirical content and limitation of IIA.

The multinomial probit model replaces the extreme-value shocks with a jointly normal vector. Its pairwise differences are generally correlated, so it does not impose IIA.

### 7.3 Random utility

**Definition 28.** Random-utility representation. Let  $A$  be a finite menu and let  $\mathbf{U} : \Omega \rightarrow \mathbb{R}^A$  be a measurable random utility vector. The induced choice probabilities are  $\rho_A(a) = \mathbb{P}\{\mathbf{U}(a) > \mathbf{U}(b) \text{ for every } b \in A \setminus \{a\}\}$ .

When  $A$  is finite and choice probabilities are observed on every nonempty menu, the deterministic types are the strict linear orders on  $A$ . Stacking the probabilities gives  $\rho = M\pi$ , where  $\pi \in \Delta(R)$  is a distribution over those orders and  $M_{(a,B),r} = \mathbb{I}\{a \text{ is } r\text{-best in } B\}$ . The finite feasibility condition  $\rho = M\pi$  with  $\pi \geq 0$  and  $\mathbf{1} \cdot \pi = 1$  is equivalent to the existence of such a distribution.

### 7.4 Random expected utility

**Definition 29.** Random expected utility. For a finite prize set  $Z$  and lotteries  $p \in \Delta(Z)$ , a random Bernoulli utility  $\mathbf{u} : \Omega \rightarrow \mathbb{R}^Z$  induces  $\mathbf{U}_\omega(p) = \sum_{z \in Z} \mathbf{u}_\omega(z)p(z)$ . Choice probabilities are obtained by maximizing this random expected utility on each menu of lotteries.

The representation imposes linearity in mixtures and regularity under menu expansion. Extremeness and weak-star continuity are additional conditions in the Gul–Pesendorfer characterization; that result is cited rather than reproved here.

### 7.5 Attributes and consumer surplus

With observed attributes  $x = (x_a)_{a \in A}$ , a common specification is  $U(a, x) = v(a, x_a) + \varepsilon_a$ , where the distribution of  $\varepsilon = (\varepsilon_a)_{a \in A}$  is independent of  $x$ . The linear-index model uses  $v(a, x_a) = x_a^\top \beta$ . Independence is a maintained exogeneity assumption, not a consequence of the random-utility definition.

**Definition 30.** Expected maximum utility. The expected value of a menu is  $V(x) = \mathbb{E}[\max_{a \in A} \{v(a, x_a) + \varepsilon_a\}]$ . If the deterministic utility contains  $-\alpha p_a$  with  $\alpha > 0$  as the price coefficient, the money-

metric consumer surplus is  $V(x)/\alpha$  up to the chosen utility normalization. If shocks are merely decision errors rather than preference shocks, this expression is not a welfare measure.

**Proposition 24.** *Williams--Daly--Zachary envelope result (cited). If the shock distribution has finite first moments and ties occur with probability zero, then  $V$  is convex and differentiable in the vector of mean utilities, and  $\frac{\partial V(x)}{\partial v(a)} = \rho(a, x)$ . Consequently the Jacobian of choice probabilities is the Hessian of  $V$ , hence symmetric and positive semidefinite. Translation invariance implies that the rows and columns sum to zero.*

The envelope identity is the basis for likelihood and welfare calculations in discrete-choice models. Additional sign restrictions such as monotone substitution patterns require additional assumptions on the shock distribution and utility index; they do not follow from random utility alone.

## Part IV

# Production and Equilibrium

## Chapter 8

# Theory of the Firm

This chapter treats technology as a feasible set of net-output vectors. Profit maximization and cost minimization are dual ways of valuing that set at prices; aggregation and efficiency follow from the same support-function geometry.

### 8.1 Production sets

There are  $L \geq 1$  commodities. A production plan  $y \in \mathbb{R}^L$  records net outputs:  $y_\ell > 0$  is an output and  $y_\ell < 0$  is an input. The production set  $Y \subseteq \mathbb{R}^L$  contains exactly the technologically feasible plans.

The benchmark assumptions are:  $Y$  is nonempty and closed;  $0 \in Y$  (inaction); and free disposal,  $y \in Y$  and  $y' \leq y$  imply  $y' \in Y$ . A no-free-lunch condition rules out  $y \in Y \cap \mathbb{R}_+^L$  other than 0. Convexity of  $Y$  is the standard diminishing-returns assumption in net-output form.

If a differentiable transformation function satisfies  $Y = \{y : F(y) \leq 0\}$  and  $F(\bar{y}) = 0$ , the local slope of the transformation frontier is

$$\frac{dy_k}{dy_\ell} = -\frac{F_\ell(\bar{y})}{F_k(\bar{y})}.$$

The minus sign is essential: increasing one net output requires a compensating reduction in the other when the frontier is written as  $F(y) = 0$ .

## 8.2 Production functions and input substitution

With  $M$  outputs and  $L - M$  inputs, write outputs as  $q \in \mathbb{R}_+^M$  and input use as  $z \in \mathbb{R}_+^{L-M}$ . In the single-output case a production function  $f$  describes free disposal through  $q \leq f(z)$ . If  $f$  is differentiable, the marginal rate of technical substitution between inputs  $\ell$  and  $k$  is

$$MRTS_{\ell k}(z) = \frac{f_\ell(z)}{f_k(z)},$$

the amount of input  $k$  needed to offset a marginal reduction in input  $\ell$  while holding output fixed. Concavity of  $f$  makes the input requirement set  $\{z : f(z) \geq q\}$  convex.

## 8.3 Profit maximization

**Definition 31.** For a price vector  $p \in \mathbb{R}^L$ , the profit function is

$$\pi(p) = \sup_{y \in Y} p \cdot y,$$

and the supply correspondence is  $y(p) = \arg \max_{y \in Y} p \cdot y$  whenever the supremum is finite and attained. If  $Y$  is closed and unbounded, finiteness and attainment require separate assumptions.

**Proposition 25.** *Profit-function properties (cited). The function  $\pi$  is convex and homogeneous of degree one in prices. The supply correspondence is homogeneous of degree zero; if  $Y$  is convex, it is convex-valued. At a differentiability point, Hotelling's lemma gives  $\nabla \pi(p) = y(p)$ . If  $\pi$  is twice differentiable,  $D_p y(p) = D_p^2 \pi(p)$  is symmetric positive semidefinite and  $D_p y(p)p = 0$ .*

The non-differentiable law of supply is the monotonicity inequality

$$(p - p') \cdot (y - y') \geq 0$$

for  $y \in y(p)$  and  $y' \in y(p')$ . It follows by adding the two profit-maximization inequalities.

For a single-output technology,  $q = f(z)$ , profit maximization is  $\max_{z \geq 0} \{pf(z) - w \cdot z\}$ . At a differentiable interior optimum,  $p \nabla f(z^*) = w$ ; with nonnegative inputs the KKT conditions are

$$pf_\ell(z^*) \leq w_\ell, \quad z_\ell^* (w_\ell - pf_\ell(z^*)) = 0.$$

## 8.4 Cost minimization

**Definition 32.** Given input prices  $w \gg 0$  and a target  $q > 0$ , the cost function is

$$c(w, q) = \min_{z \geq 0} \{w \cdot z : f(z) \geq q\}.$$

Assume the minimum is finite and attained. The conditional factor-demand correspondence is the argmin set  $z(w, q)$ . At a differentiable optimum, the KKT conditions are

$$w_\ell \geq \lambda f_\ell(z^*), \quad z_\ell^*(w_\ell - \lambda f_\ell(z^*)) = 0, \quad \lambda \geq 0.$$

**Proposition 26.** *Cost-function properties (cited). The cost function is homogeneous of degree one and concave in input prices, nondecreasing in the output target, and—when  $f$  is concave—convex in  $q$ . If the cost function is differentiable, Shephard's lemma gives  $\nabla_w c(w, q) = z(w, q)$  and  $\partial_q c(w, q) = \lambda$ . If twice differentiable,  $D_w z = D_w^2 c$  is symmetric negative semidefinite and  $D_w z w = 0$  by price homogeneity.*

The firm's output choice can be written as  $\max_{q \geq 0} \{pq - c(w, q)\}$ . If the cost function is differentiable, an interior optimum satisfies  $p = \partial_q c(w, q)$ ; at  $q = 0$  the corresponding one-sided inequality applies.

## 8.5 Aggregation and efficiency

For firms  $j = 1, \dots, J$  with production sets  $Y_j$ , the aggregate technology is the Minkowski sum

$$Y = \sum_{j=1}^J Y_j = \left\{ \sum_{j=1}^J y_j : y_j \in Y_j \right\}.$$

**Proposition 27.** *If each firm's profit maximum is finite and attained, then*

$$\pi_Y(p) = \sum_{j=1}^J \pi_{Y_j}(p), \quad y_Y(p) = \sum_{j=1}^J y_j(p)$$

where the second equality is understood as equality of attainable aggregate supply sets. The first equality follows directly from separability of the maximization over the Minkowski sum.

**Definition 33.** A feasible production plan  $y \in Y$  is efficient if there is no  $y' \in Y$  with  $y' \geq y$  and  $y' \neq y$ .

**Proposition 28.** *If  $p \gg 0$  and  $y \in y(p)$ , then  $y$  is efficient. If  $Y$  is convex and closed and satisfies free disposal, every efficient plan is supported by some nonzero  $p \geq 0$ ; this supporting-price statement is a separating-hyperplane theorem and may allow zero prices.*

## Chapter 9

# Competitive Markets and Partial Equilibrium

This chapter collects the competitive-equilibrium definitions used later and develops a transparent partial-equilibrium model with a numeraire. The partial-equilibrium model is deliberately specialized: its quasilinear preferences remove income effects for the good of interest.

### 9.1 Competitive equilibrium

There are  $I$  consumers,  $J$  firms, and  $L$  goods. Consumer  $i$  has consumption set  $X_i \subseteq \mathbb{R}^L$ , preference relation  $\succeq_i$ , and endowment  $\omega_i \in \mathbb{R}^L$ . Firm  $j$  has production set  $Y_j \subseteq \mathbb{R}^L$ , where positive coordinates are net outputs and negative coordinates are inputs. Let  $\omega = \sum_i \omega_i$ .

**Definition 34.** Feasibility. An allocation  $(x_1, \dots, x_I; y_1, \dots, y_J)$  is feasible when  $x_i \in X_i$ ,  $y_j \in Y_j$ , and

$$\sum_{i=1}^I x_i \leq \omega + \sum_{j=1}^J y_j.$$

Equality is imposed when all goods have strictly positive prices and free disposal is not used.

Competitive (Walrasian) equilibrium. A strictly positive price vector  $p \gg 0$  and a feasible allocation  $(x^*, y^*)$  form an equilibrium if every firm solves

$$y_j^* \in \arg \max_{y_j \in Y_j} p \cdot y_j$$

and every consumer solves

$$x_i^* \in \arg \max_{x_i \in X_i} \{u_i(x_i) : p \cdot x_i \leq p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*\}.$$

Ownership shares satisfy  $\theta_{ij} \geq 0$  and  $\sum_i \theta_{ij} = 1$  for each firm.

**Proposition 29.** *Price normalization and Walras' law. If  $(p, x^*, y^*)$  is a competitive equilibrium, then so is  $(\lambda p, x^*, y^*)$  for every  $\lambda > 0$ . If consumers exhaust their budgets, summing the budget equalities and using market clearing gives*

$$p \cdot \left( \sum_i x_i^* - \omega - \sum_j y_j^* \right) = 0.$$

Therefore, if all but one market clear, the last market clears as well.

## 9.2 Partial equilibrium with a numeraire

There are two goods. Good  $x$  is the good of interest and  $m$  is a numeraire with price one. Consumer  $i$  has

$$u_i(m_i, x_i) = m_i + \phi_i(x_i), \quad X_i = \mathbb{R} \times \mathbb{R}_+,$$

where  $\phi_i$  is increasing, strictly concave, twice continuously differentiable, and bounded above.

Let  $p > 0$  be the price of  $x$ .

Firm  $j$  produces  $q_j \geq 0$  units of  $x$  at numeraire cost  $c_j(q_j)$ . Its net-output set is

$$Y_j = \{(-z_j, q_j) : q_j \geq 0, z_j \geq c_j(q_j)\}.$$

Assume  $c_j$  is increasing, strictly convex, twice continuously differentiable, and  $c'_j(q) \rightarrow \infty$  as  $q \rightarrow \infty$ .

**Definition 35.** Partial competitive equilibrium. A price  $p$  and quantities  $(x_1, \dots, x_I; q_1, \dots, q_J)$  form an equilibrium when

$$\phi'_i(x_i) \leq p, \quad x_i > 0 \Rightarrow \phi'_i(x_i) = p,$$

$$p \leq c'_j(q_j), \quad q_j > 0 \Rightarrow p = c'_j(q_j),$$

and the market for  $x$  clears:

$$\sum_i x_i = \sum_j q_j.$$

These are the KKT conditions for consumer and firm optimization plus market clearing.

Equivalently, the unique choices are characterized by

$$x_i(p) = \arg \max_{x \geq 0} \{\phi_i(x) - px\}, \quad q_j(p) = \arg \max_{q \geq 0} \{pq - c_j(q)\}.$$

The associated aggregate excess demand is  $Z(p) = \sum_i x_i(p) - \sum_j q_j(p)$ .

**Proposition 30.** *Existence and uniqueness in the partial model. Suppose  $\max_i \phi'_i(0) > \min_j c'_j(0)$  and the preceding curvature and tail assumptions hold. Then  $Z$  is continuous and nonincreasing, supply is continuous and nondecreasing, and there is at least one  $p^* \in [\min_j c'_j(0), \max_i \phi'_i(0)]$  with  $Z(p^*) = 0$ . If at least one active side has strict curvature at the equilibrium, the equilibrium price is unique; otherwise the zero set can be an interval.*

To see the sign change, at a price no larger than  $\min_j c'_j(0)$  all firms choose zero output while at least one consumer demands a positive amount. At a price at least  $\max_i \phi'_i(0)$  all consumers choose zero while at least one firm supplies a positive amount. The intermediate value theorem then gives a clearing price.

The numeraire allocation is recovered from the budget identity

$$m_i = \omega_{mi} + \sum_j \theta_{ij} (pq_j - c_j(q_j)) - px_i.$$

Partial equilibrium deliberately suppresses the wealth effects that would reappear with general preferences or additional markets.

### 9.3 Welfare in the partial model

**Definition 36.** Pareto efficiency. A feasible allocation is Pareto efficient if no other feasible allocation makes every consumer weakly better off and at least one consumer strictly better off.

**Proposition 31.** *First welfare theorem for the quasilinear model (cited). A competitive allocation is Pareto efficient under the stated convexity and local nonsatiation assumptions.*

The planner's problem is

$$\max_{x_i, q_j \geq 0} \left\{ \sum_i \phi_i(x_i) - \sum_j c_j(q_j) \right\} \quad \text{subject to} \quad \sum_i x_i \leq \sum_j q_j.$$

Its KKT multiplier  $\mu$  satisfies  $\phi'_i(x_i) \leq \mu$  with equality for  $x_i > 0$  and  $\mu \leq c'_j(q_j)$  with equality for  $q_j > 0$ . Thus a competitive equilibrium satisfies the planner conditions with  $\mu = p$ .

**Proposition 32.** *Second welfare theorem for the quasilinear model (cited). Under concavity, continuity, and the usual interiority conditions, any solution of the planner's problem can be decentralized at a supporting price  $p = \mu$  by lump-sum transfers of the numeraire whose sum is zero. The theorem concerns allocations or utility vectors attainable by the planner; an arbitrary utility vector need not be feasible.*

# Chapter 10

## General Equilibrium Theory

General equilibrium studies all markets simultaneously. Prices determine individual budget sets, firms' feasible transformations, and the wealth generated by ownership claims; an equilibrium is a fixed point at which every agent optimizes and all markets clear.

### 10.1 Pure exchange economies: the Edgeworth box

Consider two consumers  $i = 1, 2$  and two goods  $\ell = 1, 2$ . Consumer  $i$  has consumption set  $X_i = \mathbb{R}_+^2$ , preference relation  $\succeq_i$ , and endowment  $\omega_i = (\omega_{1i}, \omega_{2i}) \in \mathbb{R}_+^2$ . Aggregate endowment is  $\omega = \omega_1 + \omega_2 \gg 0$ . An allocation  $x = (x_1, x_2)$  is feasible when  $x_1 + x_2 \leq \omega$  componentwise.

At prices  $p \gg 0$ , consumer  $i$  has wealth  $p \cdot \omega_i$  and budget set

$$B_i(p) = \{x_i \in X_i : p \cdot x_i \leq p \cdot \omega_i\}.$$

**Definition 37.** Walrasian equilibrium in the exchange economy. A pair  $(p^*, x^*)$  with  $p^* \gg 0$  is an equilibrium if  $x^*$  is feasible and

$$x_i^* \succeq_i x_i \quad \text{for every } x_i \in B_i(p^*), \quad i = 1, 2.$$

Prices are defined only up to multiplication by a positive scalar.

**Example 10.** Cobb–Douglas exchange. Suppose  $u_i(x_{1i}, x_{2i}) = x_{1i}^\alpha x_{2i}^{1-\alpha}$  with  $0 < \alpha < 1$ , and endowments  $\omega_1 = (1, 2)$  and  $\omega_2 = (2, 1)$ . The demands are

$$x_1(p) = \left( \frac{\alpha(p_1 + 2p_2)}{p_1}, \frac{(1 - \alpha)(p_1 + 2p_2)}{p_2} \right), \quad x_2(p) = \left( \frac{\alpha(2p_1 + p_2)}{p_1}, \frac{(1 - \alpha)(2p_1 + p_2)}{p_2} \right).$$

Clearing good 1 gives  $p_1/p_2 = \alpha/(1 - \alpha)$ ; one may normalize  $p_2 = 1$  and recover the allocation from the two demands.

**Definition 38.** Pareto efficiency. A feasible allocation  $x$  is Pareto efficient if there is no feasible  $x'$  with  $x'_i \succeq_i x_i$  for both consumers and  $x'_j \succ_j x_j$  for at least one consumer. The Pareto set is the set of all such allocations for the fixed aggregate endowment.

**Proposition 33.** *First welfare theorem for the exchange box (cited). If preferences are locally nonsatiated and  $(p^*, x^*)$  is a Walrasian equilibrium, then  $x^*$  is Pareto efficient.*

*Second welfare theorem for the exchange box (cited). If preferences are continuous, convex, and strongly monotone, every Pareto-efficient allocation can be supported by a positive price vector after suitable lump-sum transfers  $T_1 + T_2 = 0$ . The transfers change wealth, not relative prices.*

## 10.2 One consumer and one producer

There is one consumer with endowment  $\bar{L}$  of time and no initial endowment of the consumption good. Let  $x_1$  be leisure,  $x_2$  consumption, and  $f(z)$  output from labor  $z \geq 0$ . At output price  $p > 0$  and wage  $w > 0$ , the firm solves

$$\max_{z \geq 0} \{pf(z) - wz\}.$$

The consumer owns the firm, receives profit  $\pi(p, w)$ , and solves

$$\max_{(x_1, x_2) \in X} u(x_1, x_2) \quad \text{subject to} \quad px_2 \leq w(\bar{L} - x_1) + \pi(p, w).$$

**Definition 39.** A competitive equilibrium is a price pair  $(p^*, w^*)$  and choices  $(x^*, z^*)$  satisfying both optimization problems and the clearing conditions

$$x_2^* = f(z^*), \quad z^* = \bar{L} - x_1^*.$$

**Proposition 34.** *Under continuity, convexity, local nonsatiation, and the usual concavity and*

attainment assumptions, a competitive allocation solves the central-planner problem

$$\max_{x_1, x_2} u(x_1, x_2) \quad \text{subject to} \quad x_2 \leq f(\bar{L} - x_1), \quad 0 \leq x_1 \leq \bar{L}.$$

Conversely, an interior planner solution can be supported by prices equal to the relevant marginal rates.

### 10.3 Production, ownership, and feasibility

Consider  $I$  consumers,  $J$  firms, and  $L$  commodities. Consumer  $i$  has consumption set  $X_i$ , preference  $\succeq_i$ , and endowment  $\omega_i \in \mathbb{R}^L$ ; firm  $j$  has production set  $Y_j \subseteq \mathbb{R}^L$ . Let  $\omega = \sum_i \omega_i$ . An allocation  $(x, y)$  is feasible when

$$\sum_{i=1}^I x_i \leq \omega + \sum_{j=1}^J y_j.$$

In a private-ownership economy, consumer  $i$  owns share  $\theta_{ij} \in [0, 1]$  of firm  $j$ , with  $\sum_i \theta_{ij} = 1$  for each  $j$ .

**Definition 40.** Walrasian equilibrium. A price vector  $p \gg 0$  and allocation  $(x^*, y^*)$  form an equilibrium when, for every firm,

$$y_j^* \in \arg \max_{y_j \in Y_j} p \cdot y_j,$$

and, for every consumer,

$$x_i^* \in \arg \max_{x_i \in X_i} \{u_i(x_i) : p \cdot x_i \leq p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*\},$$

with market clearing  $\sum_i x_i^* = \omega + \sum_j y_j^*$ .

**Definition 41.** Price equilibrium with transfers. Replace each consumer's ownership wealth by an assigned  $w_i$  satisfying  $\sum_i w_i = p \cdot \omega + \sum_j p \cdot y_j^*$ , and require  $x_i^* \in \arg \max \{u_i(x_i) : p \cdot x_i \leq w_i\}$  while firms maximize profit and markets clear.

### 10.4 Welfare theorems

**Proposition 35.** *First Fundamental Welfare Theorem (cited).* If preferences are locally non-satiated and  $(p, x^*, y^*)$  is a price equilibrium with transfers, then  $(x^*, y^*)$  is Pareto efficient. A Walrasian equilibrium is the special case in which wealth is generated by endowments and ownership shares.

The proof is the supporting-price argument: every strictly preferred consumption bundle costs at least the consumer's assigned wealth; summing these inequalities and using firm profit maximization contradicts feasibility.

**Definition 42.** Price quasi-equilibrium with transfers. The equilibrium conditions are the same except that a bundle strictly preferred to  $x_i^*$  must have cost at least  $w_i$ ; preference maximization over the entire budget set is not required.

**Proposition 36.** *Second Fundamental Welfare Theorem (cited). Suppose every production set is convex and every preference is convex, continuous, and locally nonsatiated. Every Pareto-efficient allocation can be supported by a nonzero price vector as a price quasi-equilibrium with transfers. If assigned wealths are strictly positive and the standard boundary conditions hold, the quasi-equilibrium upgrades to a price equilibrium with transfers.*

The result follows by separating the convex aggregate set of bundles that make every consumer strictly better off from the convex aggregate production-plus-endowment set. Positivity and local nonsatiation handle the final upgrade from weak support to full budget maximization.

## 10.5 General equilibrium with inattention

An inattention model must distinguish true prices from the prices or signals used by a consumer's decision rule. Let  $M_i$  be an attention parameter and let  $x^i(p, M_i)$  be the demand correspondence generated by the fully specified behavioral model. The model must state whether  $M_i$  changes perceived prices, the budget, utility, or information.

For a price-perception benchmark,  $\tilde{p}_i = M_i p$  and a possible behavioral budget are written explicitly, for example

$$x^i(p, M_i) \in \arg \max_{x \in X_i} \{u_i(x) : \tilde{p}_i \cdot x \leq \tilde{p}_i \cdot \omega_i\}.$$

This is not a Walrasian budget unless  $M_i = I$  or the model separately establishes equivalence with the true budget.

**Definition 43.** Inattention equilibrium. A pair  $(p, x)$  is an inattention equilibrium when each  $x_i \in x^i(p, M_i)$  satisfies the model's actual feasibility restrictions and

$$\sum_{i=1}^I x_i = \sum_{i=1}^I \omega_i.$$

**Proposition 37.** *Compatibility with Pareto efficiency. Suppose preferences are differentiable, strictly increasing, and concave and  $x^*$  is an interior Pareto-efficient allocation. Then there is a common supporting price vector  $q \gg 0$  and multipliers  $\mu_i > 0$  such that  $\nabla u_i(x_i^*) = \mu_i q$  for every  $i$ . If the behavioral first-order condition is  $\nabla u_i(x_i^*) = \lambda_i M_i p$ , then each  $M_i p$  must be collinear with the same  $q$ .*

This is a necessary compatibility condition, not an if-and-only-if statement about attention matrices. Identical matrices can support efficiency in a regular model, while different matrices can also be compatible at special allocations. Perceived-price optimality alone does not justify either welfare theorem.

# Chapter 11

## Positive Theory of Equilibrium

The preceding chapter established welfare properties of competitive equilibrium. This chapter studies what those conditions imply—and what they leave undetermined—for aggregate excess demand, existence, local uniqueness, comparative statics, and the core.

### 11.1 Aggregate excess demand in pure exchange

Consider a pure-exchange economy with consumers  $i = 1, \dots, I$ , commodity space  $\mathbb{R}_+^L$ , endowments  $\omega_i \in \mathbb{R}_+^L$ , and aggregate endowment  $\omega = \sum_i \omega_i \gg 0$ . At a strictly positive price vector  $p \gg 0$ , consumer  $i$  has wealth  $w_i = p \cdot \omega_i$  and demand correspondence

$$c_i(p, w_i) = \arg \max \{u_i(x) : x \geq 0, p \cdot x \leq w_i\}.$$

**Definition 44.** Aggregate excess demand. Select a demand  $x_i(p) \in c_i(p, p \cdot \omega_i)$  and define

$$z(p) = \sum_{i=1}^I x_i(p) - \omega.$$

When demands are single-valued this is a function; with multiple optima it is a correspondence.

Pure-exchange equilibrium. A pair  $(p^*, x^*)$  with  $p^* \gg 0$  is a Walrasian equilibrium when each  $x_i^* \in c_i(p^*, p^* \cdot \omega_i)$  and  $\sum_i x_i^* = \omega$ . Prices are equivalent up to multiplication by a positive scalar.

**Proposition 38.** *Walras' law and homogeneity. Suppose each consumer is locally nonsatiated and demands are single-valued. Then*

$$p \cdot z(p) = 0, \quad z(\lambda p) = z(p) \quad (\lambda > 0).$$

If  $p \gg 0$  and markets clear in all but one commodity, Walras' law implies that the final market clears as well.

If preferences are continuous, strictly convex, and strongly monotone, the demand function is continuous on  $\mathbb{R}_{++}^L \times \mathbb{R}_{++}$ . Consequently  $z$  is continuous on strictly positive prices. Its domain should not be silently extended to a zero price: demand may become unbounded there. In an equilibrium with nonnegative prices, a good can have excess supply only at a zero price; under strong monotonicity and  $p \gg 0$ , every market clears.

For calculations it is convenient to normalize prices. We use either the simplex

$$\Delta = \{p \in \mathbb{R}_+^L : \mathbf{1} \cdot p = 1\}$$

or the chart  $p_L = 1$ . These are equivalent for strictly positive prices.

**Example 11.** Cobb--Douglas exchange. With  $u_i(x) = x_1^\alpha x_2^{1-\alpha}$  and  $0 < \alpha < 1$ ,

$$x_i(p) = \left( \frac{\alpha p \cdot \omega_i}{p_1}, \frac{(1 - \alpha) p \cdot \omega_i}{p_2} \right).$$

Substitution into  $z(p) = 0$  reduces equilibrium to one relative-price equation, because the second equation follows from Walras' law.

## 11.2 Existence

**Proposition 39.** *Walrasian existence theorem (cited). In a finite pure-exchange economy, if consumption sets are closed and convex, preferences are complete, transitive, continuous, and locally nonsatiated, and the aggregate endowment is strictly positive, a Walrasian equilibrium exists under the standard boundedness condition that makes the relevant feasible set compact. Strict convexity is not required for existence; it gives uniqueness of each consumer's demand.*

The usual proof normalizes prices to  $\Delta$ , constructs a demand or excess-demand correspondence with nonempty compact convex values, and applies a fixed-point theorem. Boundary behavior is essential: the correspondence must be defined on the closed simplex, and the assumptions must rule out a fixed point at a zero-price vector. Writing a fixed-point map only on  $\mathbb{R}_{++}^L$  does not by itself prove existence.

### 11.3 Local uniqueness and regularity

**Definition 45.** Normalized local uniqueness. A normalized equilibrium  $p^* \in \Delta$  is locally unique if some neighborhood of  $p^*$  contains no other normalized equilibrium.

Work in the chart  $p_L = 1$  and write  $\hat{z}(p_1, \dots, p_{L-1}) = (z_1, \dots, z_{L-1})$ . The last component is redundant by Walras' law.

**Definition 46.** Regular equilibrium. A differentiable equilibrium  $p^*$  is regular when the Jacobian  $D\hat{z}(p^*)$  is nonsingular.

**Proposition 40.** *Regularity implies local uniqueness (cited). If  $D\hat{z}(p^*)$  is nonsingular, the inverse function theorem implies that  $p^*$  is an isolated zero of  $\hat{z}$ . This is a local statement; it does not rule out another equilibrium far away.*

For a regular economy, the signed index of an equilibrium is the sign of the determinant of the Jacobian in an oriented price chart. The index theorem (cited) states that, under the usual compactness and boundary hypotheses, the sum of indices over normalized equilibria equals one. Hence a regular economy has an odd number of equilibria. The conclusion is global and requires the theorem's boundary assumptions; it does not follow from the inverse function theorem alone.

### 11.4 What equilibrium theory does not identify

At a differentiability point, Walras' law and homogeneity imply the exact derivative restrictions

$$Dz(p)^\top p = -z(p), \quad Dz(p)p = 0.$$

At an equilibrium, the first identity becomes  $Dz(p)^\top p = 0$ ; these are restrictions, not a complete theory of comparative statics.

**Proposition 41.** *Sonnenschein--Mantel--Debreu theorem (cited). Subject to continuity, homogeneity of degree zero, Walras' law, and the standard boundary condition, aggregate excess demand can have essentially arbitrary shape on compact subsets of the strictly positive normalized price simplex. In particular, individual rationality does not generally imply uniqueness, stability, or monotone comparative statics for the aggregate equilibrium map.*

The theorem is an impossibility result about restrictions on aggregate demand. It does not say that every arbitrary function is a demand system: the stated regularity, Walras, homogeneity,

and boundary conditions remain necessary, and the result concerns a constructed finite exchange economy.

## 11.5 Uniqueness and gross substitutes

With a constant-returns production cone  $Y \subseteq \mathbb{R}^L$  containing zero, a strictly positive price  $p$  supports the technology when  $p \cdot y \leq 0$  for every  $y \in Y$ . If aggregate excess demand satisfies  $z(p) \in Y$  and Walras' law, then choosing production  $y = z(p)$  clears markets and maximizes profit. This observation separates the consumer condition from the technology-support condition.

**Definition 47.** Gross substitutes. An excess-demand function has the strict gross-substitutes property if, whenever only the price of good  $\ell$  rises, excess demand for every other good  $k \neq \ell$  strictly rises. Weak gross substitutes replaces strict inequalities with weak inequalities. The property is imposed on the relevant price domain and is stronger than Walras' law or homogeneity.

**Proposition 42.** *Gross-substitutes uniqueness theorem (cited). Under continuity, homogeneity, Walras' law, and a strict gross-substitutes condition, a pure-exchange equilibrium has at most one normalized strictly positive price vector. Existence still requires a separate existence theorem. Without strictness, gross substitutes can deliver a convex or lattice-structured equilibrium set rather than a unique price.*

## 11.6 The core

For clarity, define the core first in a pure-exchange economy. There are  $I$  consumers, endowments  $\omega_i \in \mathbb{R}_+^L$ , and a feasible allocation satisfies  $\sum_i x_i = \sum_i \omega_i$  (free disposal can replace equality by  $\leq$ ). A nonempty coalition  $S \subseteq \{1, \dots, I\}$  can use only  $\sum_{i \in S} \omega_i$ .

**Definition 48.** Blocking coalition. Coalition  $S$  blocks a feasible allocation  $x$  if there are  $\{x'_i\}_{i \in S}$  with  $x'_i \succ_i x_i$  for every  $i \in S$  and  $\sum_{i \in S} x'_i \leq \sum_{i \in S} \omega_i$ .

The core is the set of feasible allocations blocked by no coalition.

**Proposition 43.** *Walrasian allocations are in the core (cited). Suppose preferences are locally nonsatiated and  $(p^*, x^*)$  is a Walrasian equilibrium with  $p^* \gg 0$ . If a coalition made every member strictly better off, each member would need a bundle with strictly greater value than  $p^* \cdot \omega_i$ . Summing over the coalition contradicts its resource constraint. Thus no coalition blocks  $x^*$ .*

The converse need not hold in a finite economy: the core can contain noncompetitive allocations. In replica economies, the core-convergence theorem (cited) says that as the number

of identical copies grows, core allocations approach the set of Walrasian allocations under the standard convexity, continuity, and nonsatiation assumptions.

## Part V

# Advanced Equilibrium Applications

## Chapter 12

# Testable Restrictions on the Equilibrium Manifold

This chapter studies what a finite set of market observations can reveal about a pure-trade economy. The central tension is between individual utility maximization and market clearing: each consumer must rationalize their own choices, while the choices must add up to the observed aggregate endowment.

### 12.1 Observables and the equilibrium manifold

Fix a finite set of consumers  $i = 1, \dots, I$ , goods  $\ell = 1, \dots, L$ , and observations  $t = 1, \dots, T$ . At observation  $t$ , let  $p^t \in \mathbb{R}_{++}^L$  denote the price vector and let  $\omega_i^t \in \mathbb{R}_+^L$  denote consumer  $i$ 's endowment. Prices are normalized because only their positive scale is economically relevant.

**Definition 49.** (Pure-trade rationalization.) A finite dataset is rationalized by a pure-trade economy if there are continuous, locally nonsatiated, concave utility functions  $u_i$  and bundles  $x_i^t \in \mathbb{R}_+^L$  such that, for every  $i, t$ ,  $x_i^t \in \arg \max\{u_i(x) : p^t \cdot x \leq p^t \cdot \omega_i^t\}$ , and, for every  $t$ ,  $\sum_i x_i^t = \sum_i \omega_i^t$ .

(Equilibrium manifold.) For a market excess-demand function  $F(\hat{\omega}, p)$ , where  $\hat{\omega} = (\omega_1, \dots, \omega_I)$ , the equilibrium manifold is  $\mathcal{M} = \{(\hat{\omega}, p) : F(\hat{\omega}, p) = 0\}$ . It records pairs of endowment profiles and prices that can clear markets for the fixed underlying preferences.

## 12.2 Afriat inequalities and elimination of unobservables

For a finite collection of observed or latent bundles, Afriat's inequalities are the linear inequalities  $u_i^t \leq u_i^s + \lambda_i^s p^s \cdot (x_i^t - x_i^s)$  for all  $s, t$ , with  $\lambda_i^s > 0$ . They are equivalent to the existence of a monotone, concave utility representation for the finite choice data.

The equilibrium restrictions combine these inequalities with the budget inequalities  $p^t \cdot x_i^t \leq p^t \cdot \omega_i^t$  and the market-clearing equalities  $\sum_i x_i^t = \sum_i \omega_i^t$ . The  $x_i^t, u_i^t, \lambda_i^t$  variables are unobserved; prices and the endowment profile are the data in the design considered here.

(Tarski--Seidenberg theorem.) The projection of a semialgebraic set is semialgebraic. Consequently, eliminating the unobserved variables from any finite polynomial system yields a quantifier-free semialgebraic condition in the observables. This theorem supplies an elimination principle; it does not say that the resulting restrictions are easy to compute or that every model has a nontrivial restriction.

## 12.3 Brown--Matzkin theorem

Brown and Matzkin (1996) apply this principle to pure-trade equilibrium. Under their regularity and invariance assumptions, finite observations on prices, individual incomes or endowments, and aggregate endowments are rationalizable if and only if they satisfy a finite family of semialgebraic restrictions obtained after eliminating the latent utility levels, marginal utilities of income, and consumption bundles. The restrictions are nontrivial: some finite datasets cannot lie on the equilibrium manifold of any economy in the maintained class. The result is cited here; its quantifier-elimination construction is not reproved in this text.

The theorem should not be confused with the Sonnenschein--Mantel--Debreu result. The latter concerns the flexibility of aggregate excess demand when individual preferences are allowed to vary; Brown--Matzkin hold the individual preferences fixed across observations and exploit observed endowment changes.

# Chapter 13

## Sequential Trade

### 13.1 Arrow--Debreu benchmark

There are  $S$  states and  $L$  goods in each state. Consumer  $i$  has state-contingent endowment  $\omega_i = (\omega_{i1}, \dots, \omega_{iS})$  and preference  $\succeq_i$  over contingent consumption plans  $x_i = (x_{i1}, \dots, x_{iS})$ .

**Definition 50.** (Arrow--Debreu equilibrium.) Prices  $p = (p_1, \dots, p_S) \in (\mathbb{R}_{++}^L)^S$  and an allocation  $x^* = (x_1^*, \dots, x_I^*)$  form an Arrow--Debreu equilibrium if, for each consumer,  $x_i^*$  maximizes  $\succeq_i$  over  $B_i^{AD}(p) = \{x_i : \sum_{s=1}^S p_s \cdot x_{is} \leq \sum_{s=1}^S p_s \cdot \omega_{is}\}$ , and markets clear state by state:  $\sum_i x_{is}^* = \sum_i \omega_{is}$  for every  $s$ .

### 13.2 Spot markets and sequential trade

Suppose contingent commodity markets are unavailable at date  $t = 0$ . There are  $K$  tradable assets with payoff matrix  $R \in \mathbb{R}^{S \times K}$ : one unit of asset  $k$  pays  $R_{sk}$  units of the first good in state  $s$ . Asset prices are  $q \in \mathbb{R}^K$ , and spot-good prices in state  $s$  are  $p_s \in \mathbb{R}_{++}^L$ .

**Definition 51.** (Radner budget set.) Given asset holdings  $z_i \in \mathbb{R}^K$  and state-contingent consumption  $x_i$ , consumer  $i$  satisfies  $q \cdot z_i \leq 0$  and  $p_s \cdot x_{is} \leq p_s \cdot \omega_{is} + p_{s1}(Rz_i)_s$  for every  $s$ . The zero on the date-0 constraint is the normalization of zero initial asset endowment.

(Radner equilibrium.) A price system  $(q, p_1, \dots, p_S)$  and plans  $(z_i^*, x_i^*)_{i=1}^I$  form a Radner equilibrium when every consumer maximizes their preference over the Radner budget set, asset markets clear  $\sum_i z_i^* = 0$ , and goods markets clear in every state:  $\sum_i x_{is}^* = \sum_i \omega_{is}$  for all  $s$ .

**Proposition 44.** (Arrow--Radner equivalence, cited.) If the payoff matrix has full row rank  $\text{rank}(R) = S$  and the standard no-arbitrage, continuity, and monotonicity assumptions hold, the

*sets of consumption allocations supported by Arrow--Debreu and Radner equilibria coincide. If  $\text{rank}(R) < S$ , some contingent trades are unattainable and the equivalence need not hold.*

The full-rank condition is the precise meaning of complete asset markets in this finite-state model. The equivalence theorem is cited rather than reproved; the proof constructs state prices from asset prices and uses the spanning property of  $R$ .

# Chapter 14

## Matching

### 14.1 One-to-one matching

Let  $M$  and  $W$  be finite sets of agents. Each agent has a strict preference ordering over the agents on the other side and the option of remaining unmatched. A matching  $\mu$  assigns each agent either an acceptable partner or themselves, and satisfies  $\mu(\mu(a)) = a$ .

**Definition 52.** (Stability.) A matching is individually rational if no agent is matched with an unacceptable partner. A pair  $(m, w)$  not matched to one another blocks  $\mu$  if  $w \succ_m \mu(m)$  and  $m \succ_w \mu(w)$ . A matching is stable when it is individually rational and has no blocking pair.

### 14.2 Deferred acceptance

In the men-proposing deferred-acceptance algorithm, each unmatched man proposes to the most-preferred woman who has not rejected him. Each woman tentatively holds her most-preferred acceptable proposal received so far and rejects the rest. The process stops when no proposal is rejected.

**Theorem 10.** (*Gale--Shapley theorem, cited.*) *For every finite marriage problem, deferred acceptance terminates and returns a stable matching. Among all stable matchings, the outcome is weakly best for every man and weakly worst for every woman.*

Termination is finite because a man proposes to each woman at most once. The stability and proposer-optimality statements are the Gale--Shapley theorem; they are cited here rather than accompanied by an incomplete proof.

### 14.3 Housing markets and top trading cycles

In a housing market, each agent initially owns one house and has a strict preference over houses. The core consists of allocations for which no coalition can make every member weakly better off and at least one member strictly better off using only the houses owned by that coalition.

The top-trading-cycles algorithm directs each agent to point to their favorite remaining house and each house to point to its owner. Every directed cycle is executed, the participating houses and agents are removed, and the procedure is repeated.

**Theorem 11.** (*Top-trading-cycles theorem, cited.*) *With strict preferences and one-house ownership, top trading cycles terminates, is Pareto efficient, and returns an allocation in the core of the housing market.*

## Chapter 15

# Computable General Equilibrium

### 15.1 From accounts to counterfactuals

Computable general equilibrium (CGE) analysis uses a calibrated microeconomic model and a social accounting matrix to evaluate policy counterfactuals. Calibration chooses parameters so that the model reproduces a benchmark allocation and the associated accounting identities before the policy is changed.

### 15.2 A two-sector small open economy

The benchmark economy has two goods, labor  $\ell$ , and sector-specific capital  $K_i$ . A small open economy takes world prices as given. For sector  $i$ , a constant-elasticity-of-substitution production function is  $Q_i = A_i[\alpha_i K_i^{-\rho_i} + (1 - \alpha_i)\ell_i^{-\rho_i}]^{-1/\rho_i}$ , with  $A_i > 0$ ,  $0 < \alpha_i < 1$ , and  $\rho_i > -1$ ,  $\rho_i \neq 0$ . The associated elasticity parameter is  $\sigma_i = 1/(1 + \rho_i) > 0$ .

**Definition 53.** (Cost and conditional factor demand.) Given output  $\bar{Q}_i$  and factor prices  $(r_i, W)$ , the unit-consistent cost function is  $c_i(r_i, W, \bar{Q}_i) = \min_{K_i, \ell_i \geq 0} \{r_i K_i + W \ell_i : Q_i(K_i, \ell_i) \geq \bar{Q}_i\}$ . When differentiable, conditional factor demands satisfy  $K_i^D = \partial c_i / \partial r_i$  and  $\ell_i^D = \partial c_i / \partial W$ . This derivative formulation avoids confusing the price  $P_i$  with the CES parameter  $\rho_i$ .

### 15.3 Social accounting and calibration

The social accounting matrix records production, factor payments, household income, and expenditure flows that must balance at the benchmark. For a Cobb–Douglas expenditure

equation  $P_i Q_{h,i}^D = \beta_{h,i} Y_h$ , the observed benchmark share identifies  $\beta_{h,i} = P_i Q_{h,i}^D / Y_h$ . The calibrated model is checked by substituting the parameter values back into every accounting identity and market-clearing condition.

## 15.4 Counterfactual policies

For an imported good  $i$ , let  $ER$  be the exchange rate and  $PW_i$  the world price. A tariff rate  $\tau_i \geq 0$  changes the domestic price according to  $P_i = ER \cdot PW_i(1 + \tau_i)$ . Free trade is the baseline  $\tau_i = 0$ ; a tariff counterfactual changes  $\tau_i$  and resolves the full equilibrium system. Other counterfactuals change tax, subsidy, transfer, or technology equations while leaving the accounting identities and calibrated preference and technology parameters in place.

# Appendix A

## Order Theory

This mathematical appendix collects the order-theoretic concepts used throughout the book. The orientation is economic:  $xRy$  means that  $x$  is ranked at least as high as  $y$ . Unless stated otherwise,  $X$  is a nonempty set.

### A.1 Binary Relations

**Definition 54.** A binary relation on  $X$  is a subset  $R \subseteq X \times X$ . We write  $xRy$  in place of  $(x, y) \in R$ . The inverse relation  $R^{-1}$  is defined by  $xR^{-1}y \iff yRx$ .

#### A.1.1 Properties

A relation  $R$  on  $X$  is:

**Reflexive** if  $xRx$  for every  $x \in X$ .

**Complete** (or total) if, for every  $x, y \in X$ , either  $xRy$  or  $yRx$  (or both). Completeness implies reflexivity by taking  $x = y$ .

**Symmetric** if  $xRy$  implies  $yRx$ .

**Asymmetric** if  $xRy$  implies  $\neg(yRx)$ . Every asymmetric relation is irreflexive.

**Antisymmetric** if  $xRy$  and  $yRx$  imply  $x = y$ . Antisymmetry does not mean the negation of symmetry.

**Transitive** if  $xRy$  and  $yRz$  imply  $xRz$ .

**Acyclic** if there do not exist  $n \geq 1$  and  $x_1, \dots, x_n \in X$  such that  $x_1 R x_2, \dots, x_{n-1} R x_n$  and  $x_n R x_1$ . Acyclicity is normally imposed on a strict relation.

**Example 12.** The equality relation on  $X$  is reflexive, symmetric, transitive, and antisymmetric, but it is incomplete whenever  $X$  contains two distinct elements.

### A.1.2 Important Classes of Relations

**Definition 55.** An equivalence relation is reflexive, symmetric, and transitive. A preorder is reflexive and transitive. A partial order is reflexive, transitive, and antisymmetric. A total preorder is a complete preorder. A linear order (or total order) is a complete partial order. A set equipped with a partial order is a partially ordered set, or poset.

In consumer theory, a rational weak preference relation is modeled as a total preorder. Antisymmetry is generally inappropriate for weak preference because distinct alternatives may be indifferent.

**Example 13.** The usual relation  $\geq$  is a linear order on  $\mathbb{R}$ . The componentwise relation on  $\mathbb{R}^L$ , defined by  $x R y \iff x_\ell \geq y_\ell$  for every  $\ell$ , is a partial order but is not complete when  $L \geq 2$ . For example,  $(1, 2)$  and  $(2, 1)$  are incomparable.

The lexicographic relation on  $\mathbb{R}^2$  is defined by  $x R_{\text{lex}} y$  if  $x_1 > y_1$  or if  $x_1 = y_1$  and  $x_2 \geq y_2$ . It is a linear order. Although complete and transitive, it has no continuous real-valued representation on  $\mathbb{R}^2$  with its usual topology.

## A.2 Symmetric and Asymmetric Parts

**Definition 56.** The symmetric part  $I_R$  and asymmetric part  $P_R$  of a relation  $R$  are defined by

$$x I_R y \iff x R y \text{ and } y R x, \quad x P_R y \iff x R y \text{ and } \neg(y R x).$$

**Proposition 45.** If  $R$  is a preorder, then  $I_R$  is an equivalence relation and  $P_R$  is asymmetric and transitive.

*Proof.* Reflexivity and symmetry of  $I_R$  follow immediately, and its transitivity follows from transitivity of  $R$ . Asymmetry of  $P_R$  is built into its definition. If  $x P_R y$  and  $y P_R z$ , then transitivity gives  $x R z$ . If  $z R x$ , then  $z R y$ , contradicting  $y P_R z$ . Thus  $x P_R z$ .  $\square$

### A.3 Equivalence Classes and Quotient Orders

**Definition 57.** Let  $E$  be an equivalence relation. The equivalence class of  $x$  is  $[x]_E = \{y \in X : yEx\}$ . The quotient set is  $X/E = \{[x]_E : x \in X\}$ .

**Example 14.** If  $X = \{a, b, c\}$  and the only distinct equivalent elements are  $a$  and  $b$ , then  $X/E = \{\{a, b\}, \{c\}\}$ . For a preference preorder, the equivalence classes of  $I_R$  are indifference classes.

**Proposition 46.** Let  $R$  be a preorder and let  $I_R$  be its symmetric part. On  $X/I_R$ , define

$$[x]_{I_R} \bar{R} [y]_{I_R} \iff xRy.$$

Then  $\bar{R}$  is well defined and is a partial order.

*Proof.* Suppose  $xI_R x'$ ,  $yI_R y'$ , and  $xRy$ . Then  $x'RxRyRy'$ , so transitivity gives  $x'Ry'$ ; hence the definition is independent of representatives. Reflexivity and transitivity are inherited from  $R$ . If both  $[x]_{I_R} \bar{R} [y]_{I_R}$  and  $[y]_{I_R} \bar{R} [x]_{I_R}$ , then  $xI_R y$ , so  $[x]_{I_R} = [y]_{I_R}$ . Thus  $\bar{R}$  is antisymmetric.  $\square$

### A.4 Maximal and Greatest Elements

**Definition 58.** For a nonempty subset  $S \subseteq X$ , the set of **maximal elements** is

$$\text{Max}(S, R) = \{x \in S : \nexists y \in S \text{ such that } yP_R x\}.$$

The set of **greatest elements** is

$$\text{Great}(S, R) = \{x \in S : xRy \text{ for every } y \in S\}.$$

A greatest element dominates every feasible alternative; a maximal element is merely undominated. Every greatest element is maximal. If  $R$  is complete and transitive on  $S$ , then the two notions coincide. Under a partial order there is at most one greatest element, while there may be many maximal elements. Under a preorder, several indifferent alternatives may all be greatest.

**Example 15.** Let  $S = \{a, b, c\}$  and suppose the only strict comparison is  $aP_R b$ . Then  $a$  and  $c$  are maximal, but there is no greatest element because  $a$  and  $c$  are incomparable.

**Proposition 47.** Every nonempty finite subset of a set equipped with a total preorder has at least one greatest element and therefore at least one maximal element.

*Proof.* Proceed by induction on the cardinality of  $S$ . The claim is immediate for a singleton. Suppose  $g$  is greatest in a finite set  $S$ , and add  $a \notin S$ . Completeness gives either  $aRg$  or  $gRa$ . In the first case, transitivity and  $gRy$  for every  $y \in S$  imply that  $a$  is greatest in  $S \cup \{a\}$ . In the second case,  $g$  remains greatest.  $\square$

Finiteness matters. For example,  $(\mathbb{R}, \geq)$  is linearly ordered but has no greatest element.

## A.5 Transitive Closure

**Definition 59.** Let  $R^1 = R$ , and recursively define  $R^{n+1} = R^n \circ R$ . The transitive closure of  $R$  is

$$R^+ = \bigcup_{n \geq 1} R^n.$$

The reflexive-transitive closure is  $R^* = \Delta_X \cup R^+$ , where  $\Delta_X = \{(x, x) : x \in X\}$ .

**Proposition 48.** *The relation  $R^+$  is the smallest transitive relation containing  $R$ .*

*Proof.* Concatenating an  $R$ -path of length  $m$  with one of length  $n$  yields a path of length  $m + n$ , so  $R^+$  is transitive. Any transitive relation containing  $R$  must contain every  $R^n$  and hence their union.  $\square$

## A.6 Order Extensions

**Definition 60.** Let  $R$  and  $R'$  be preorders on  $X$ . We call  $R'$  an extension of  $R$  if  $R \subseteq R'$  and  $P_R \subseteq P_{R'}$ . Thus every original weak and strict comparison is preserved.

**Theorem 12.** (*Szpilrajn extension theorem*) *Every partial order on a set has a linear-order extension.*

*Proof.* The standard proof applies Zorn's lemma to the set of partial orders extending the given order, ordered by inclusion. The union of every chain is again a partial order, so a maximal extension exists. If two distinct elements were incomparable in that extension, one could consistently add a comparison between them and take the transitive closure, contradicting maximality. Hence the maximal extension is complete and therefore linear.  $\square$

**Corollary 2.** *Every preorder has a total-preorder extension.*

*Proof.* Let  $I_R$  be the symmetric part of the preorder  $R$ . The induced relation  $\bar{R}$  on  $X/I_R$  is a partial order by the earlier proposition. Next, let  $\bar{R}^{\text{ext}}$  be a linear-order extension of  $\bar{R}$ , and

define on  $X$

$$xR^{\text{ext}}y \iff [x]_{I_R} \bar{R}^{\text{ext}} [y]_{I_R}.$$

Then  $R^{\text{ext}}$  is a total preorder. It contains  $R$ , and antisymmetry of  $\bar{R}^{\text{ext}}$  ensures that every comparison in  $P_R$  remains strict.  $\square$

## A.7 Checklist for Economic Applications

- Use a total preorder for rational weak preference; use its asymmetric part for strict preference and its symmetric part for indifference.
- Do not call a strict relation a partial order: strict relations are irreflexive, whereas partial orders are reflexive.
- Distinguish maximal from greatest alternatives whenever the relation may be incomplete.
- Use quotient notation  $X/I_R$ , not set-difference notation, when collapsing indifference classes.
- State finiteness, compactness, continuity, or another attainment condition whenever existence of a maximal or greatest alternative is claimed.